

Probability, Statistics & Data Analysis

5: Normal Distribution, Statistics and Intervals

**I think it is much more interesting to
live with uncertainty than to live with
answers that might be wrong.**

Richard Feynman

Lecture 5: Normal Distribution, Statistics and Intervals

What we will cover

- Normal Distribution
 - Standard Normal Distribution
 - Numbers of sigma
- Definition of a Statistic
- Central Limit Theorem
- Point Estimators
- Precision of the sample mean
 - SE, ME and CI
- Precision of a future value
 - PI

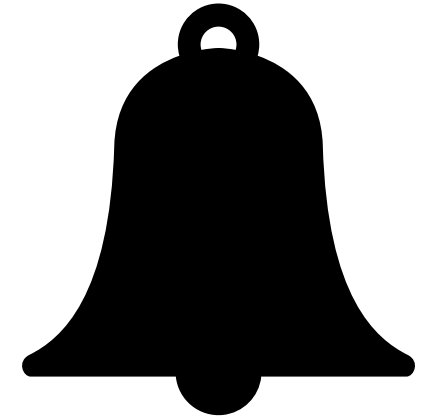




The Normal Distribution

The Normal Distribution

- Before we talk about statistics, we must discuss the ironically named Normal distribution (it is quite special, as we will see!)
- AKA
 - Gaussian
 - Bell-curve
- The expected value and variance are $E(X) = \mu$, $V(X) = \sigma^2$
- Where μ and σ^2 are the formulae we used in Lecture 1

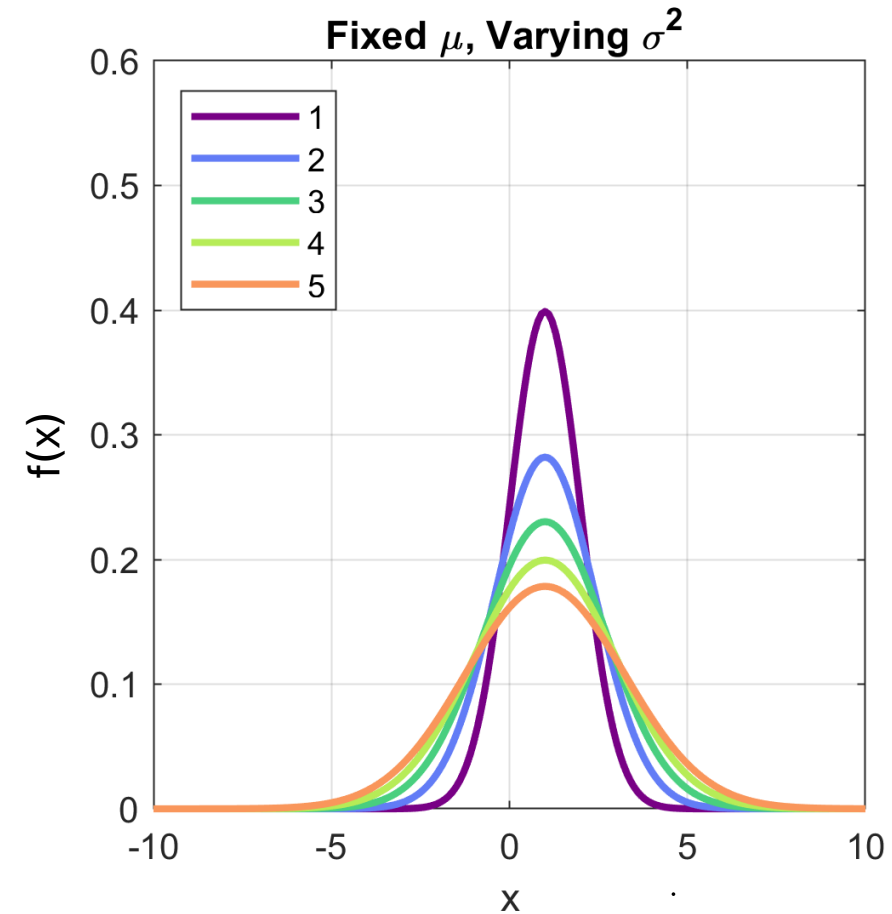
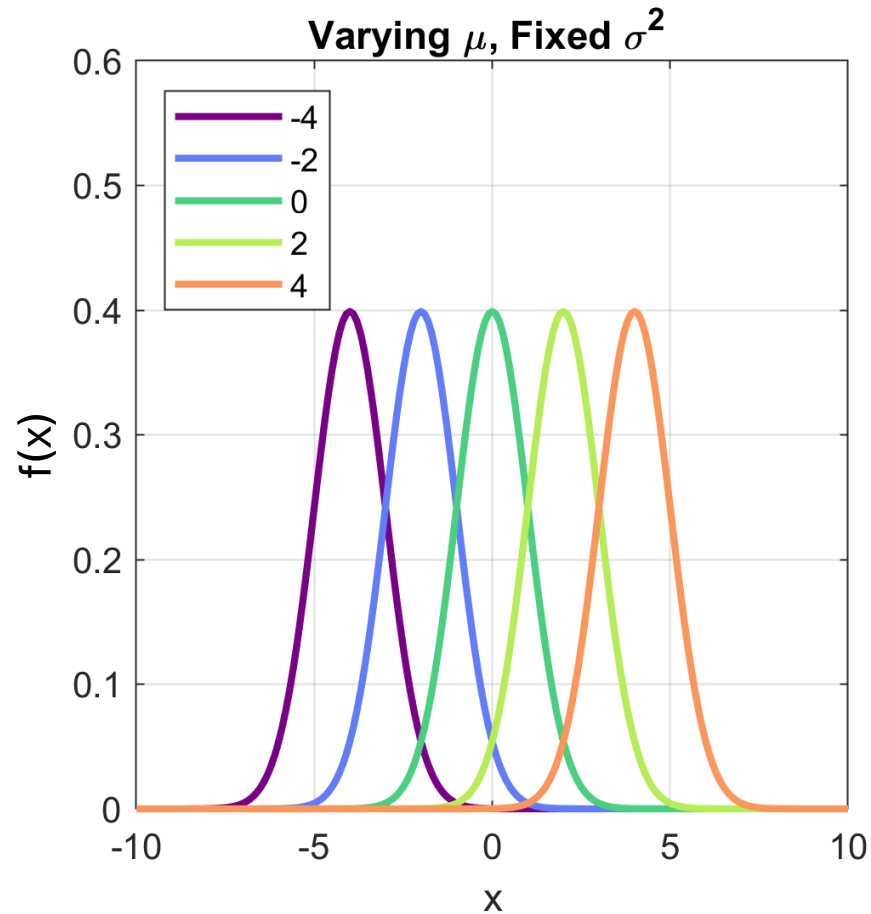


$$\mu = \frac{1}{N} \sum_{i=1}^N x_i$$

$$X \sim N(\mu, \sigma^2)$$
$$E(X) = \mu, V(X) = \sigma^2$$

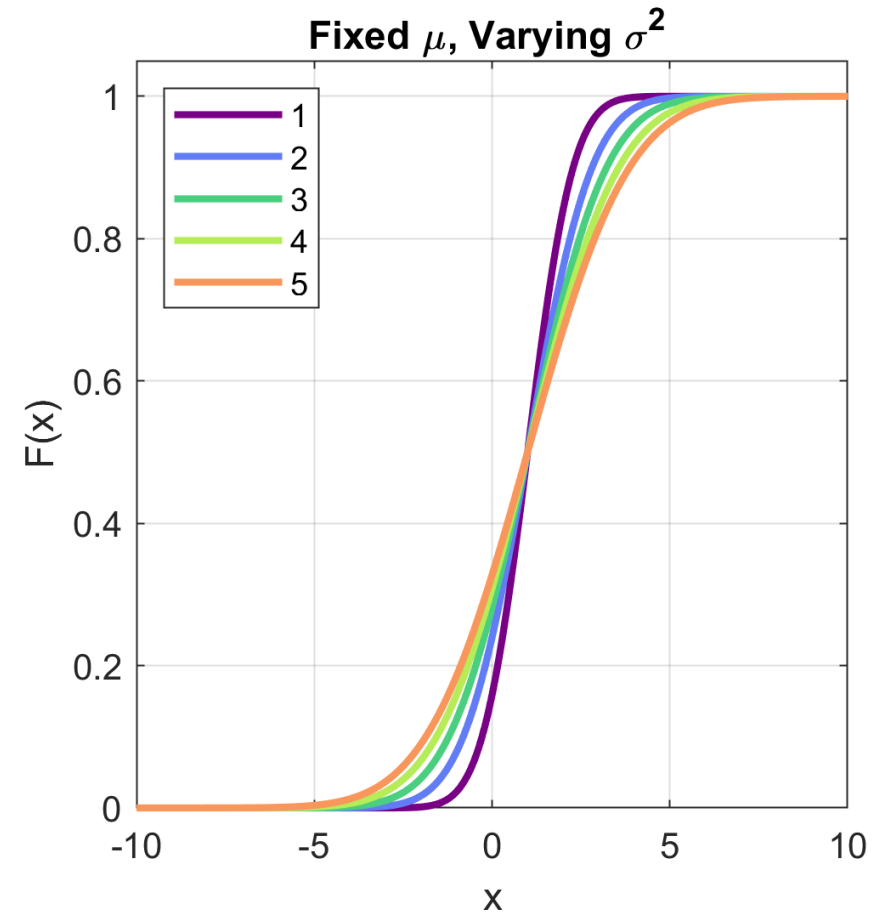
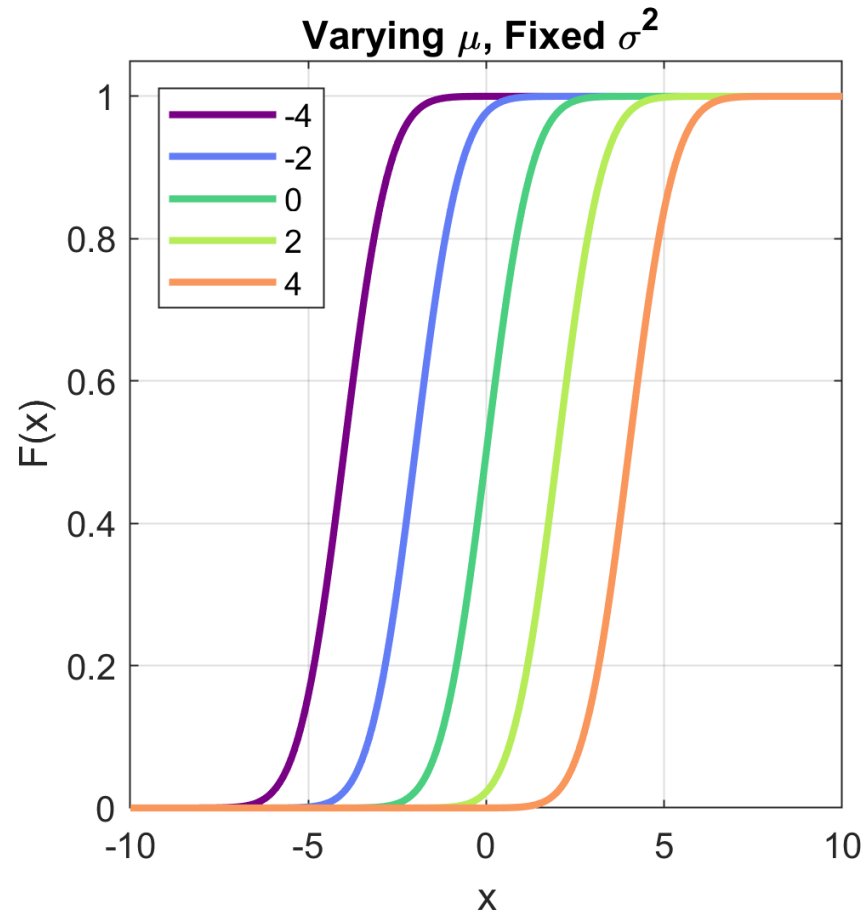
$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2$$

Normal Distribution PDF



$$f(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Normal Distribution CDF



$$F(x; \mu, \sigma^2) = \frac{1}{2} \left[1 + \operatorname{erf} \left(\frac{x - \mu}{\sqrt{2\sigma^2}} \right) \right] \quad \operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$$



The width, W , of red blood cells is described by a normal distribution with $\mu = 8 \mu m$ and $\sigma^2 = 0.5 \mu m^2$

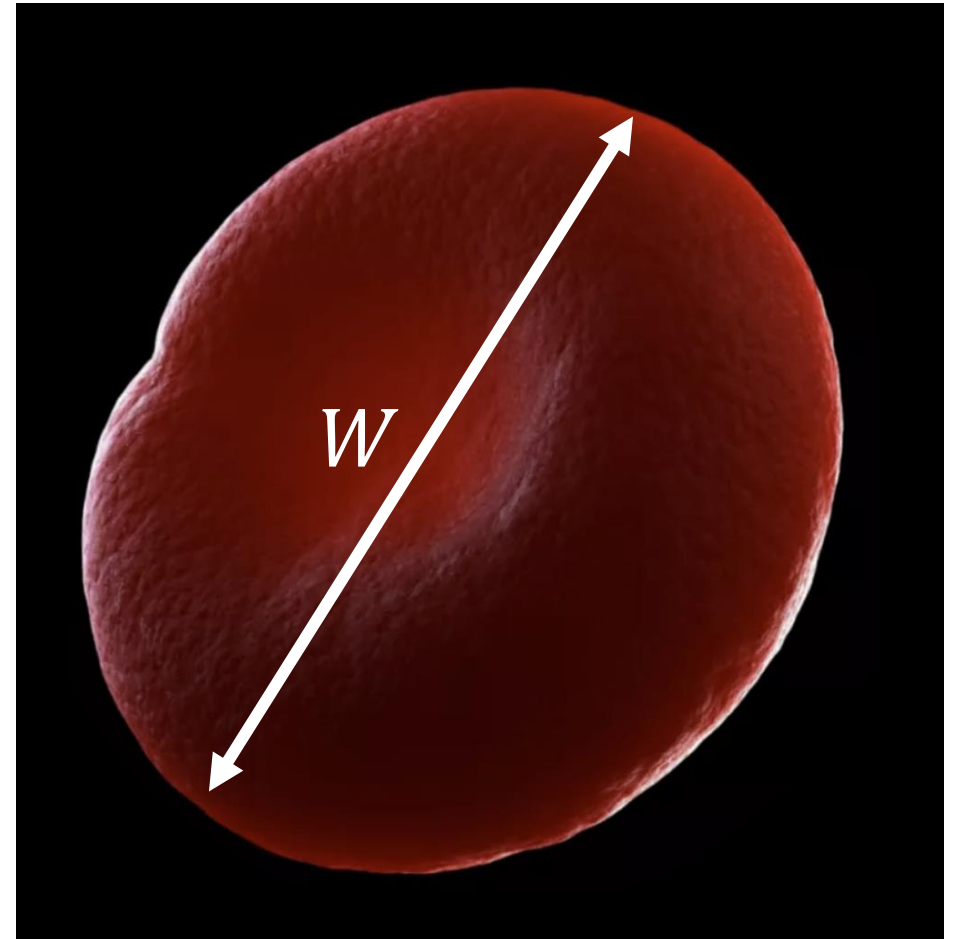
$$f(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$f(w) = \begin{cases} \frac{1}{\sqrt{\pi}} e^{-(w-8)^2} & w > 0 \\ 0 & \text{otherwise} \end{cases}$$

Find the upper and lower limits that bound 95% percent of the cells lie (i.e. the 2.5th and 97.5th percentile)

$$F(x; \mu, \sigma) = \frac{1}{2} \left[1 + \operatorname{erf} \left(\frac{x - \mu}{\sqrt{2\sigma^2}} \right) \right]$$

$$P(W \leq a) = F(a) = \frac{1}{2} [1 + \operatorname{erf}(a - 8)]$$



Normal Distribution Probability- RBC width



- $\mu = 8 \mu m$ and $\sigma^2 = 0.5 \mu m^2$

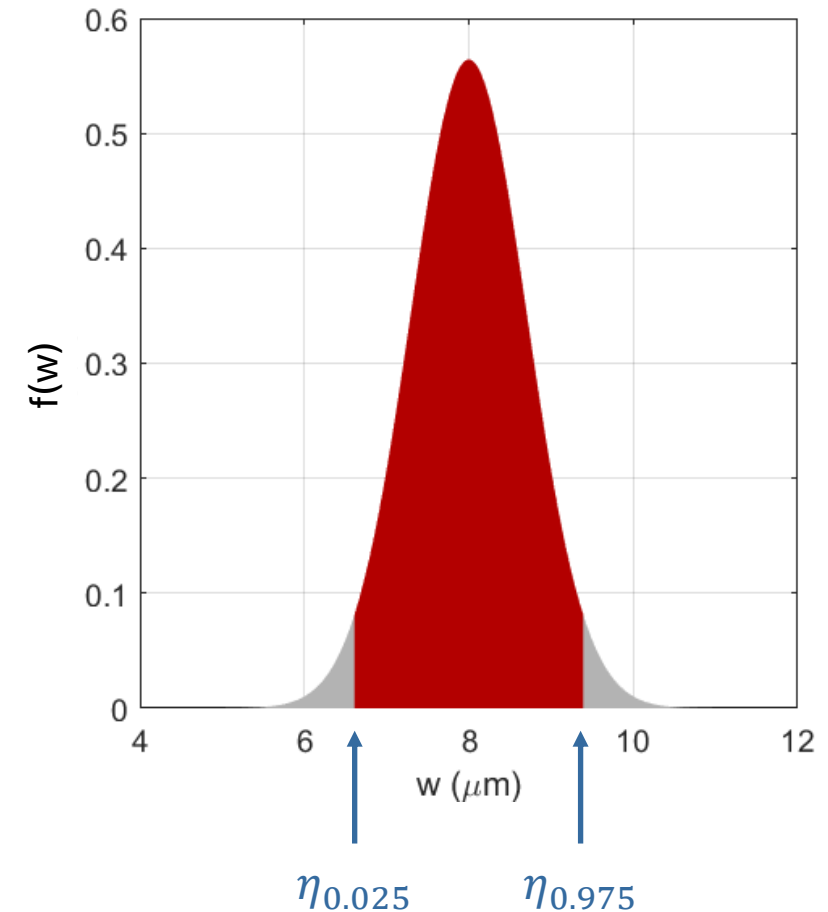
$$P(W \leq a) = F(a) = \frac{1}{2} [1 + \operatorname{erf}\left(\frac{a - 8}{\sqrt{2\sigma^2}}\right)]$$

$$F(x; \mu, \sigma) = \frac{1}{2} \left[1 + \operatorname{erf}\left(\frac{x - \mu}{\sqrt{2\sigma^2}}\right) \right]$$

```
>> erfinv(-0.95)
ans =
-1.3859
```

```
>> erfinv(0.95)
ans =
1.3859
```

- Easy enough with MATLAB to do the inverse erf
- Without a computer, less so





The standard normal distribution

A normalised normal distribution

Normalising the Normal Distribution

- In general, finding the probability that a normally distributed variable will be within a certain range is used a lot
- For this we need to compute

$$P(a \leq X \leq b) = \int_a^b \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dy$$

or via CDF

$$P(a \leq X \leq b) = \frac{1}{2} \left[1 + \operatorname{erf} \left(\frac{b - \mu}{\sqrt{2\sigma^2}} \right) \right] - \frac{1}{2} \left[1 + \operatorname{erf} \left(\frac{a - \mu}{\sqrt{2\sigma^2}} \right) \right]$$

- Pretty unwieldy!
- A little easier is the special case for $\mu = 0$ and $\sigma = 1$
- We can generalise this outcome by defining

$$Z = \frac{X - \mu}{\sigma}$$

Note σ is selected to be the standard deviation of the variable X

Standard Normal Distribution

$\Phi(z) = F(z)$ for
standard normal

$$Z = \frac{X - \mu_X}{\sigma_X}, \quad Z \sim N(0,1)$$

$$\mu_Z = 0 \quad \sigma_Z = 1$$

$$f(X; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$F(X; \mu, \sigma^2) = \frac{1}{2} \left[1 + \operatorname{erf} \left(\frac{x - \mu}{\sqrt{2\sigma^2}} \right) \right]$$

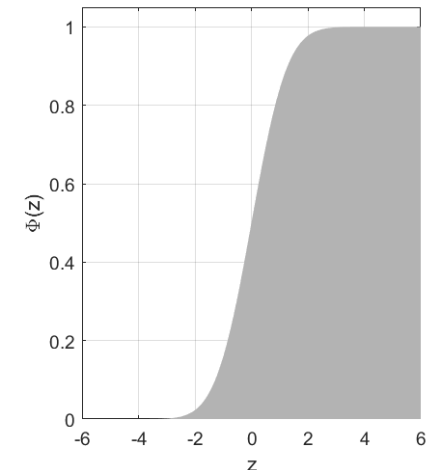
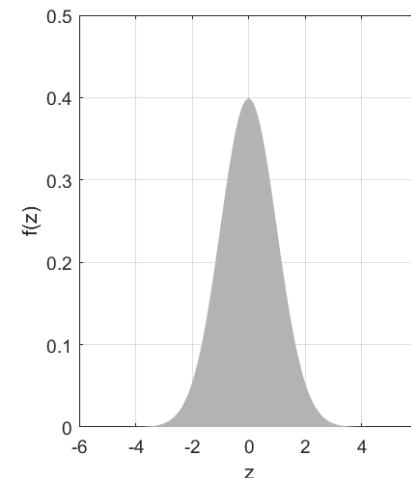
$$f(z; 0,1) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}$$

$$\Phi(z) = F(z; 0,1) = \frac{1}{2} \left[1 + \operatorname{erf} \left(\frac{z}{\sqrt{2}} \right) \right]$$

**PDF of Standard
normal distribution**

**CDF of Standard
normal distribution**

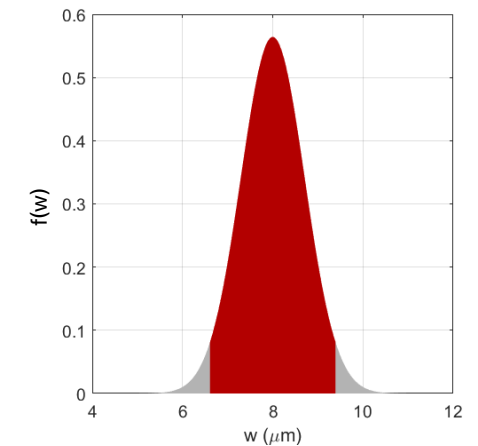
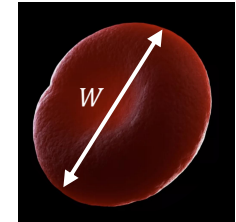
- Still not trivial, but we can use stats tables for $\Phi(z)$ and remember certain values, e.g.
 - $\Phi(1.96) = 0.975$ (you'll see why later)
- Any normally distributed variable can be converted to the standard normal distribution ('standardised') for analysis



Using the Standard Normal Distribution – RBC width



- The width, W , of red blood cells is described by a normal distribution with $\mu = 8 \mu\text{m}$ and variance $\sigma^2 = 0.5 \mu\text{m}^2$ (standard deviation $\sigma = \sqrt{0.5 \mu\text{m}^2} = \frac{1}{\sqrt{2}} \mu\text{m}$)
- To get to $\mu = 0 \mu\text{m}$ and $\sigma = 1 \mu\text{m}$



Using Φ table – RBC width



$$\Phi[z] = 0.975$$

$$z = 1.9 + 0.06 = 1.96$$

```
>> norminv(0.975)

ans =

    1.9600
```

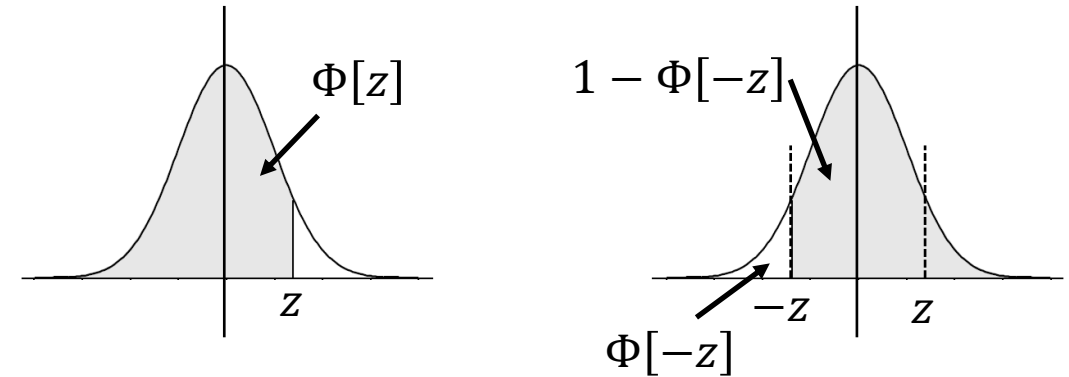
		Second decimal place of z							
		0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07
First decimal place of z	0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279
	0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675
	0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064
	0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443
	0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808
	0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157
	0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486
	0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794
	0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078
	0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340
	1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577
	1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790
	1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980
	1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147
	1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292
	1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418
	1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525
	1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616
	1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693
	1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756
	2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808
	2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850

RBC width for $\Phi[z] < 0.5$



Next is $\Phi[z] = 0.025$

- Some tables only provide $\Phi[z] > 0.5$
- However, $\Phi[-z] = 1 - \Phi[z]$



$1 - \Phi[-z] = \Phi[z]$ due to symmetry

Or use MATLAB

```
>> norminv(0.025)

ans =

-1.9600
```



Important Features of Normal Distribution

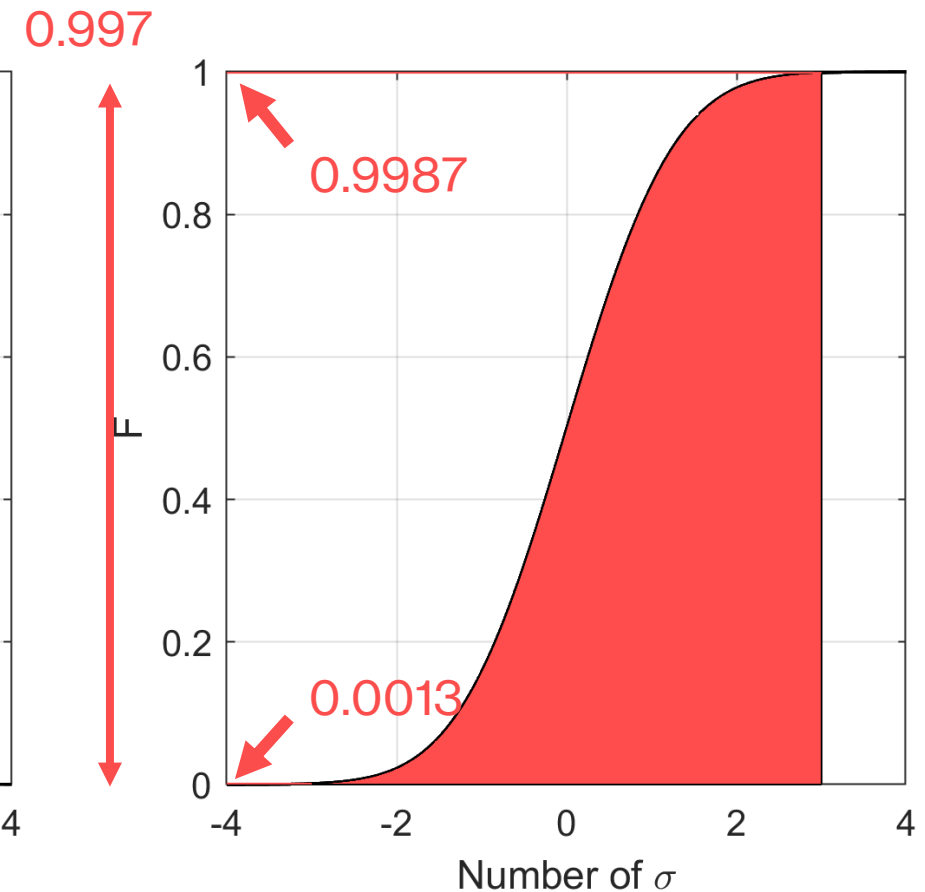
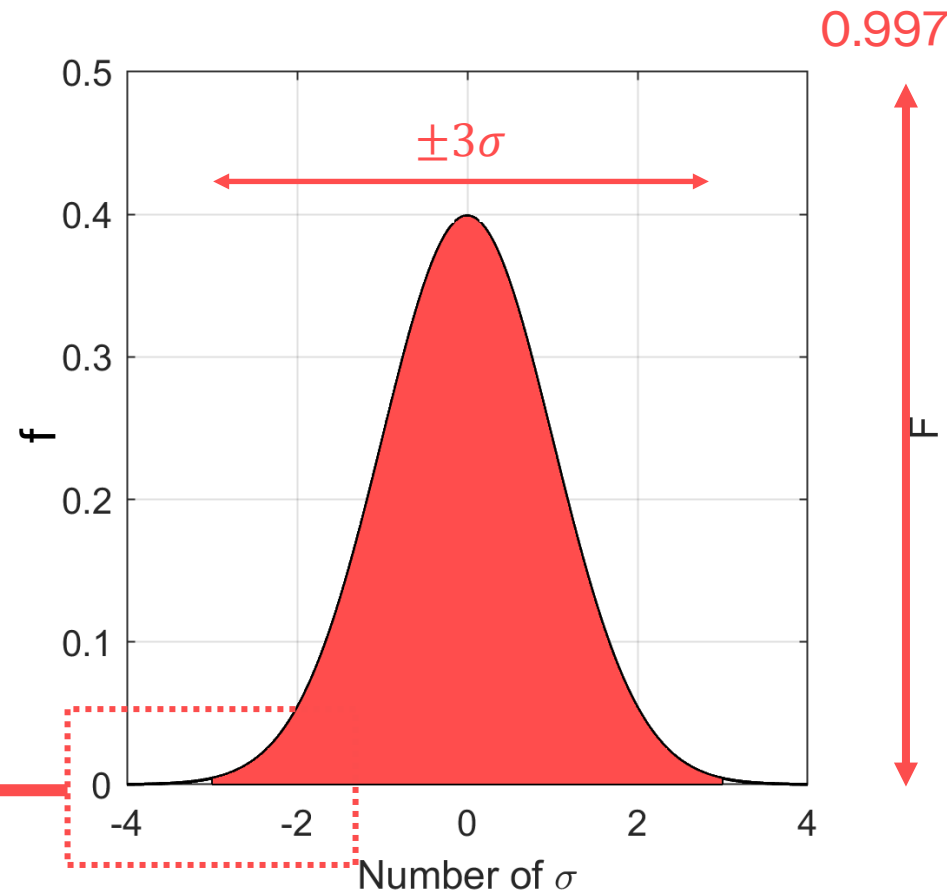
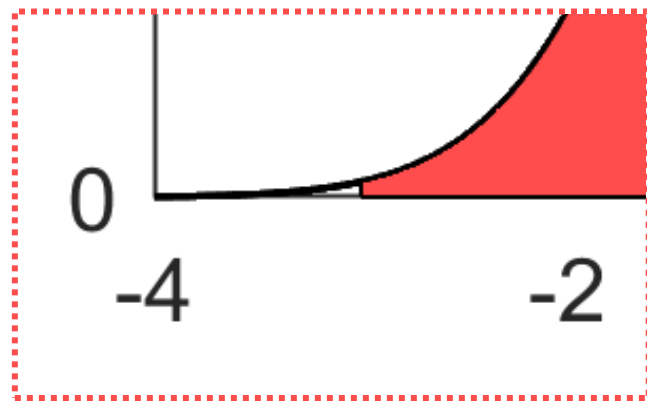
Numbers of Sigma and Combinations

Numbers of sigma: 3σ

- Standardisation of normal distribution also provides useful reference frame for probabilities
- z axis can be interpreted as numbers of standard deviations
- For $\pm 3\sigma$

$$p(|Z| \leq 3\sigma) \approx 99.7\%$$

$$= \Phi(3) - \Phi(-3)$$

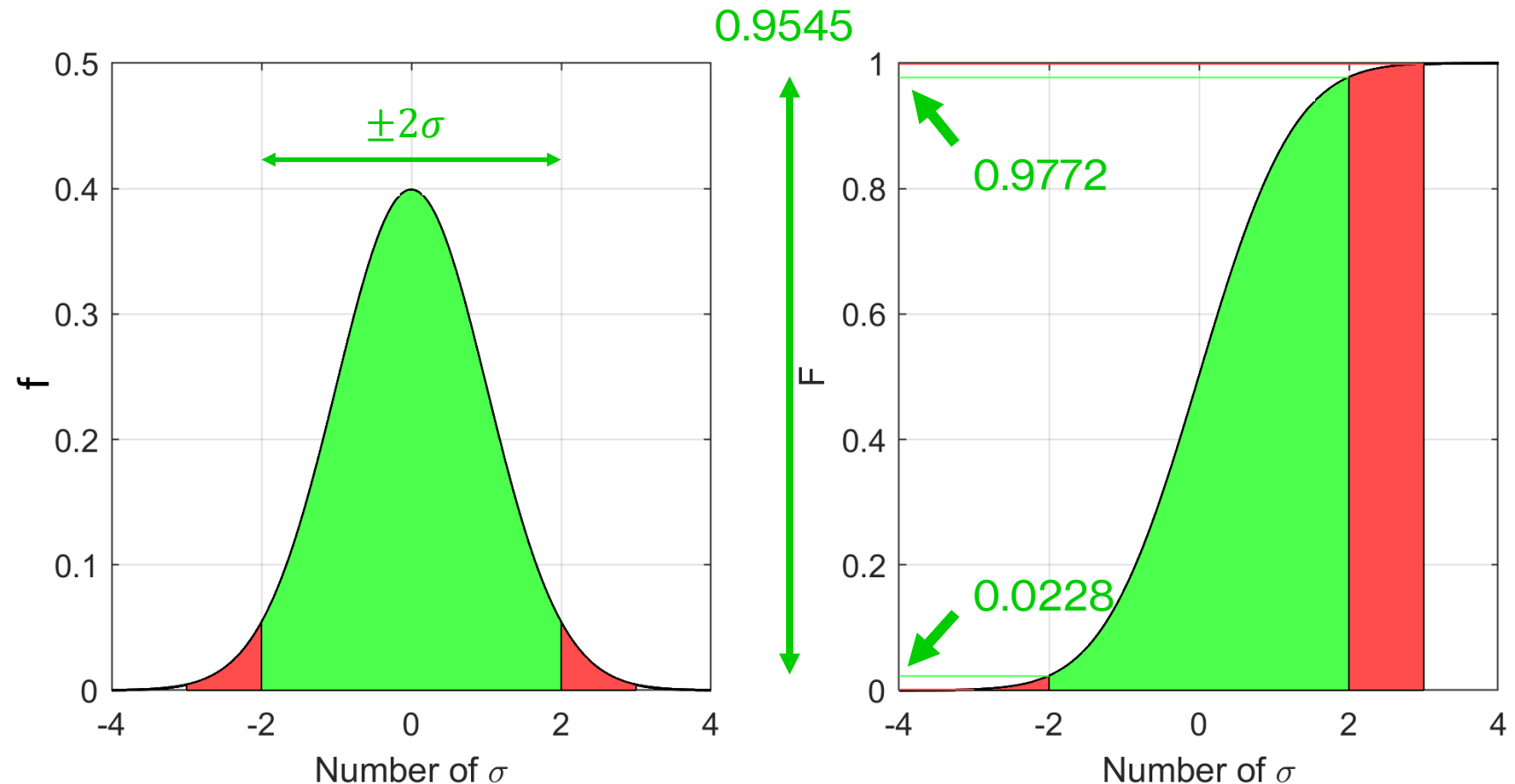


Numbers of sigma: 2σ

- Standardisation of normal distribution also provides useful reference frame for probabilities
- For $\pm 2\sigma$

$$p(|Z| \leq 2\sigma) \approx 95\%$$

$$= \Phi(2) - \Phi(-2)$$



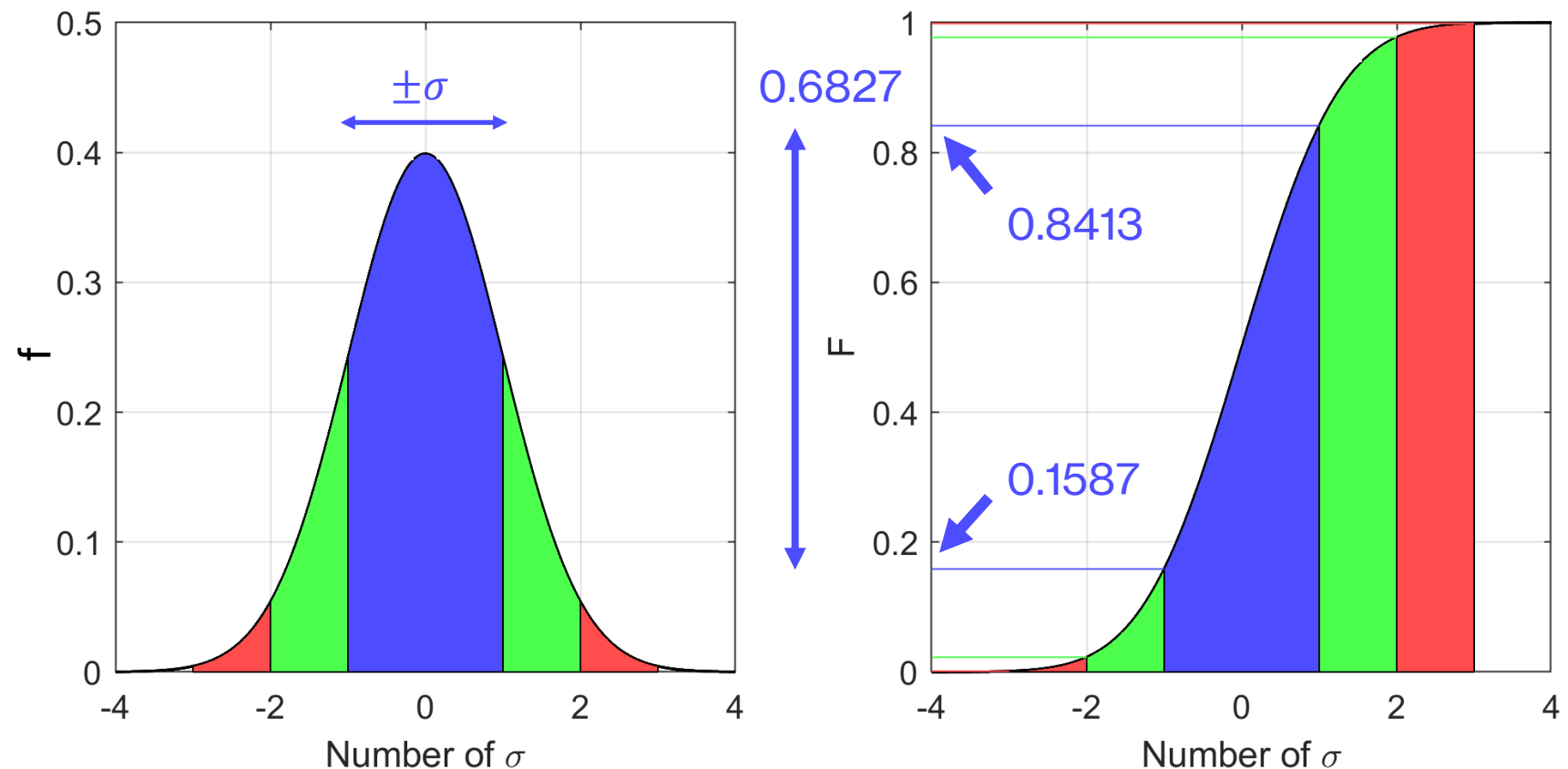
Numbers of sigma: 1σ

- Standardisation of normal distribution also provides useful reference frame for probabilities
- For $\pm 1\sigma$

$$p(|Z| \leq \sigma) \approx 68\%$$

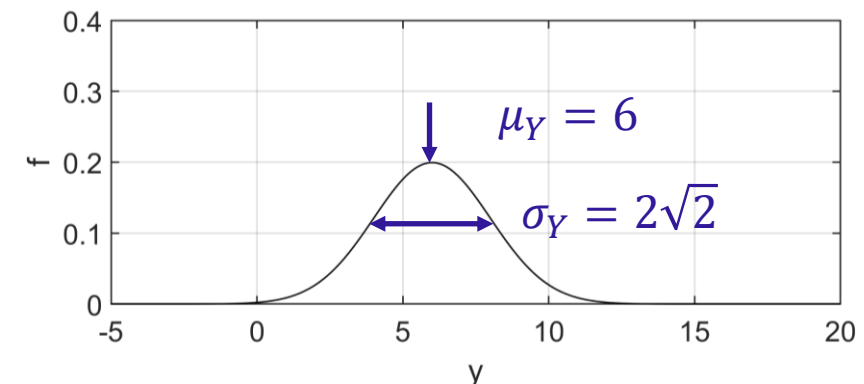
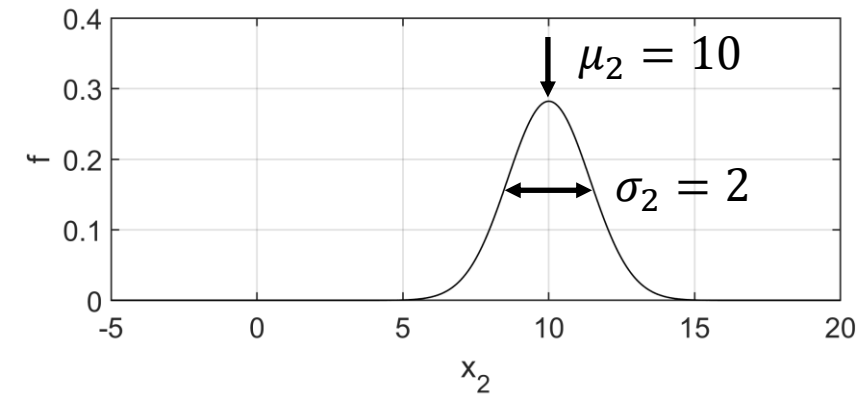
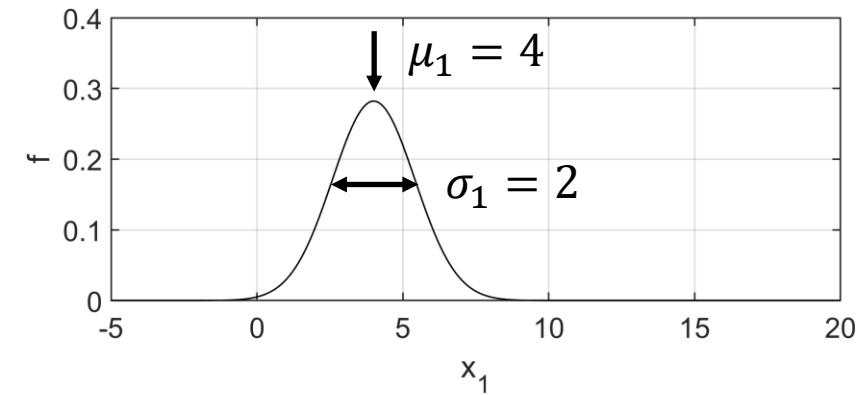
$$= \Phi(1) - \Phi(-1)$$

Why this is important will become clear in due course



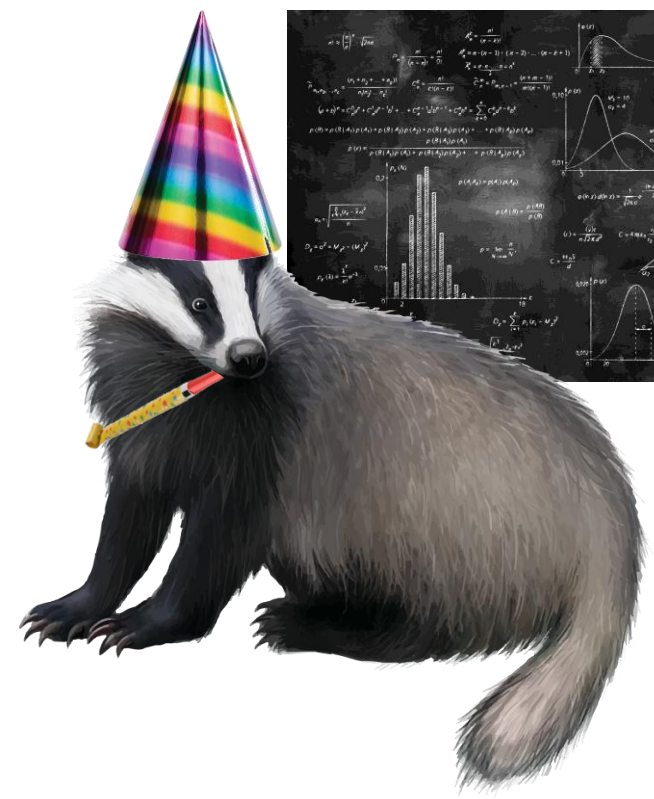
Linear Combinations of Normal Distributions

- Let $X_1 \sim N(\mu_1, \sigma_1^2)$ and $X_2 \sim N(\mu_2, \sigma_2^2)$
- Then if $Y = X_2 - X_1$
- Expected value/Mean
 - $E(Y) = E(X_2 - X_1) = E(X_2) - E(X_1)$
 - $\mu_Y = \mu_{X_2} - \mu_{X_1}$
 - The means subtract
- Variance
 - $V(Y) = V(X_2 - X_1) = V(X_2) + V(X_1)$
 - $\sigma_Y^2 = \sigma_{X_2}^2 + \sigma_{X_1}^2$
 - The variances add
 - The uncertainties (SD) add ‘in quadrature’
 - $\sigma_Y = \sqrt{\sigma_{X_2}^2 + \sigma_{X_1}^2}$



Note for addition, means add, variances add

“A badger celebrating statistics”
DeepAI



“A badger celebrating statistics”
My attempt

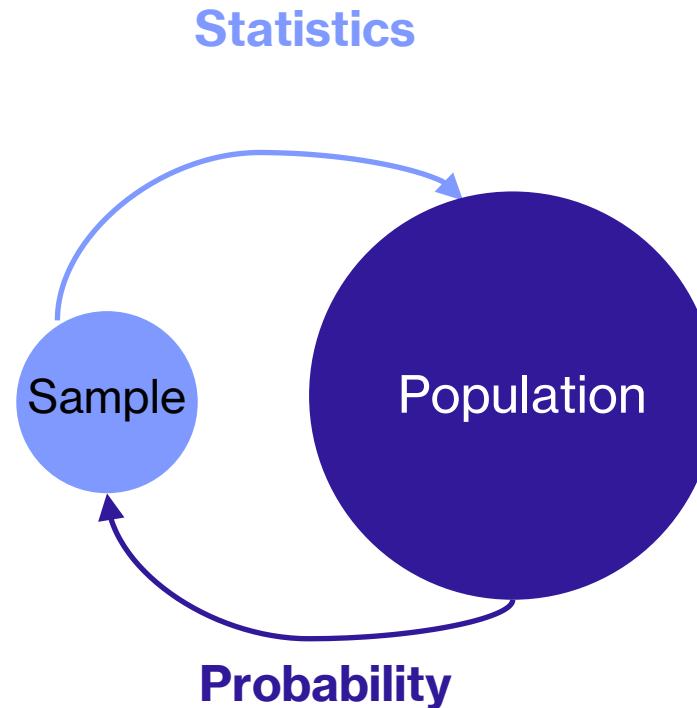
Statistics

What actually is one?

Probability and Statistics

Inferential Statistics

- We use a sample to estimate something about the population
- We posit that the population has a specific distribution
 - Often the normal distribution
- We check if this is a good model
- If so, we use our knowledge of probability to calculate how likely it was that we observed this outcome



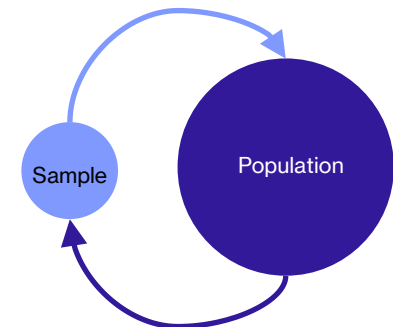
Probability

- We already know the details of the population
- We want to calculate how likely a certain outcome is for a given sample
- Mostly, we use probability to understand the foundations of statistics



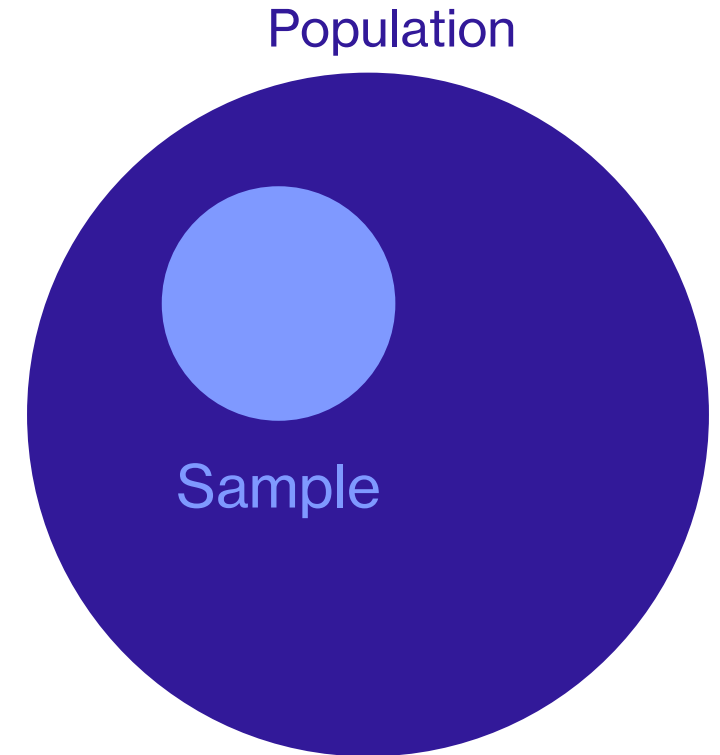
Quick return to RVs

- RV: a rule that associates a number with an outcome in the sample space, $\mathcal{S}$
 - X is the random variable, it is random because before the experiment we don't know what value will occur
 - x is an observed value of X , once the experiment has been carried out
- When we know $p(x) = \dots$ or $f(x) = \dots$ then we can use probability to predict future outcome
- Typically, we don't know $p(x)$ or $f(x)$ but instead have data
 - $x = [1, 5, 7, 9, 2, 4, 6, 8]$
 - We can use descriptive statistics to characterise this sample of x
 - But the next sample of x might be different!
 - Need to understand the uncertainty of a given sample
- Inferential Statistics



Formal Definition of a Statistic

- A statistic is a quantity that can be calculated from sample data
 - E.g. average RBC diameter
 - E.g. median absolute deviation of white blood cell count following a treatment
 - E.g. interquartile range of bone density in osteoporotic femur
- Prior to taking a sample, there is uncertainty as to what the resulting statistic will be
- A statistic is therefore a **Random Variable**



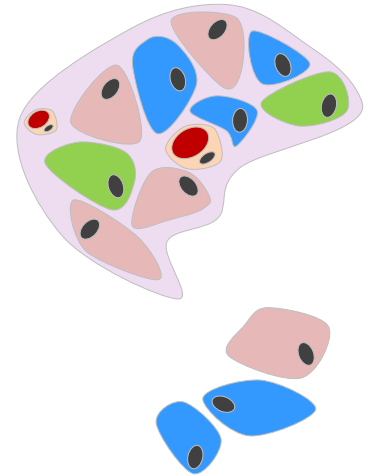
Random Samples and Capitalisation

Random Sample

RVs $X_1, X_2, \dots, X_n$ form a random sample if

- The individual values of X_i are independent
 - With replacement or
 - from infinite population or
 - ~ from small sample relative to population (e.g. 20x)
- Every X_i has the same probability distribution

This would
not be a good
example of
random
sampling

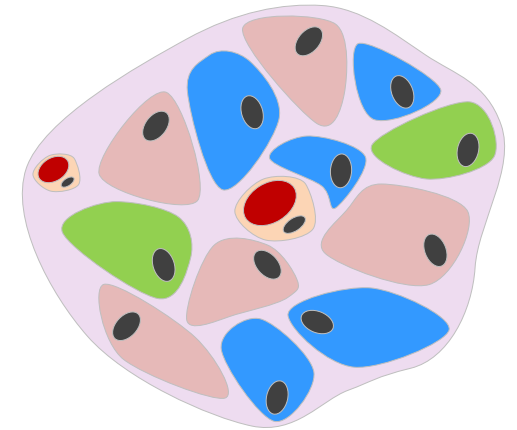


Capitalisation of statistics

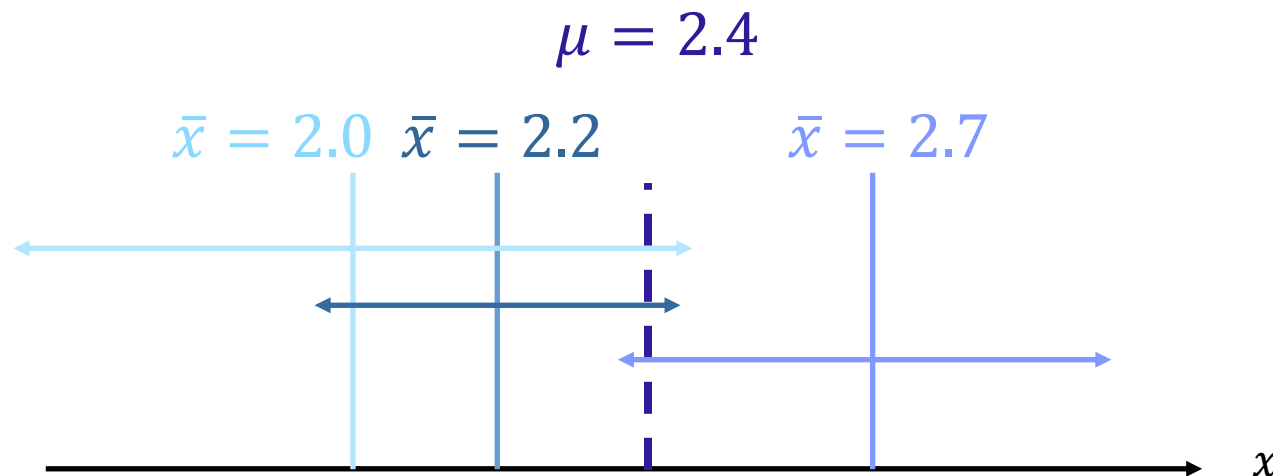
- Before an experiment, an observation is a **random variable** X because it could take on one of many values
- After an experiment, we have an **observation** x_i , because it is a single realisation of X_i
- This applies also to sample statistics such as $\bar{X}$ and S : a single realisation of which gives $\bar{x}$ and s

Cell heterogeneity

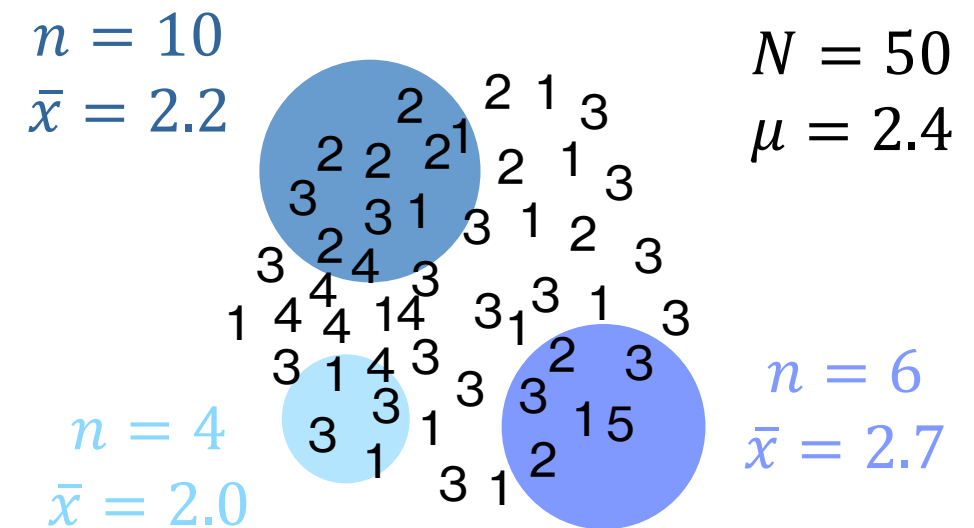
- Let our statistic be the sample mean of X
- Define an RV: $\bar{X}$
 - Lower case $\bar{x}$ is a realisation of $\bar{X}$
- We calculate $\bar{x}$ in order to estimate μ



Pancreatic islet



- We want to quantify the likelihood that $\bar{x} \pm ?$ includes the population mean μ



These are technically too small to be random samples



Monte Carlo Simulations

Simulating the process of random sampling



- The role of sampling is not always intuitive
- Monte Carlo simulation is a numerical method that can help with this
- It can also help us to find some important solutions without needing to complex derivations
- For our purposes, the process is
 - Start with a known population distribution, e.g. $X \sim N(\mu, \sigma^2)$ or $X \sim \text{Pois}(\lambda)$
 - Simulate a large number, B , of samples (e.g. $B = 100$), with each sample containing n_{obs} observations
 - Change n_{obs} and repeat
 - Evaluate how parameters change as the sample size changes
- Use this to learn something about what happens in the ‘real-world’ wherein B is typically 1

Monte Carlo Casino



I borrowed the notation B from bootstrapping, a different numerical method

N is more commonly used, but more prone to ambiguity

Badger Weights

- A wildlife conservationist wants to know the average weight of the badgers living in a certain area
- They cannot measure all the local badgers, so instead will gather a single sample of badgers (with n_{obs} badgers) and weigh them to get an estimate of the population average weight
- How will the variability of the measurement of average weight be affected by the number of badgers weighed?
- We will use Monte Carlo simulations to investigate



Sampling Simulation – Badger Weights



- Let X be the weight of a badger in kg
- The real-world scenario is
 - We want to know μ , but we cannot measure all badgers, so need to take samples and find $\bar{x}_i$
- To simulate, we assume a specific distribution
 - $X \sim N(10, 4)$
- Each sample contains `n_obs` observations (often called sample size)
- We will simulate $B = 100$ samples with `n_obs`=5, 10, 20, 50 and 100
- For each value of `n_obs` we evaluate the mean and standard deviation of the 100 values of $\bar{x}_i$

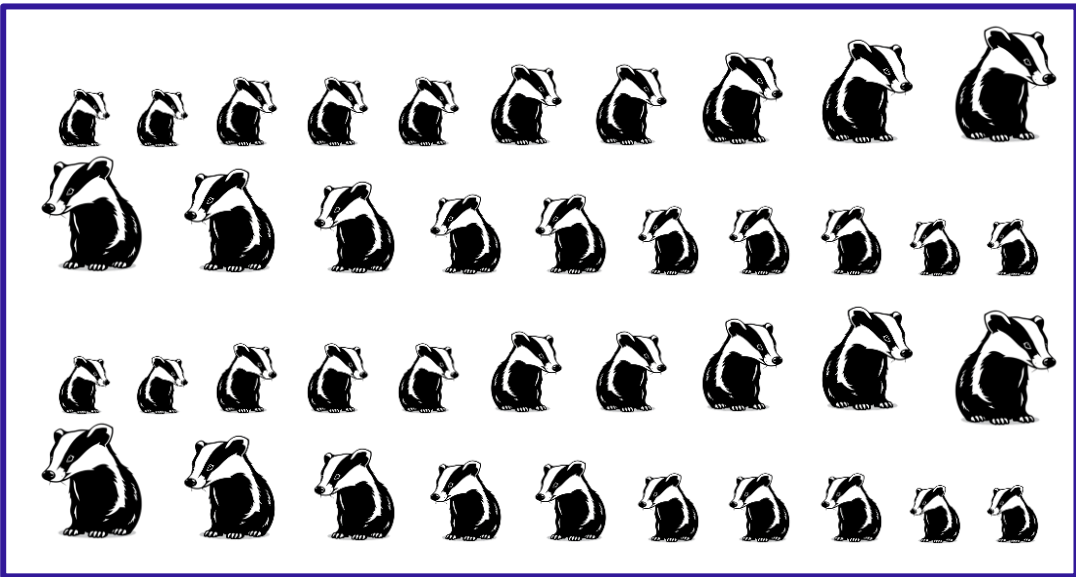


Sampling Simulation – Badger Weights



The badger population

deepai.org/machine-learning-model/text2img



- Start with $n_{obs} = 5$
- Simulate the sample
 $xi=randn(n_obs)*sigma+mu$
- From B simulated samples, calculate $\bar{y}$ and s_Y : the mean and standard deviation of the sample means

$\bar{x}_i = y$

$$\bar{y} = \frac{1}{B} \sum_{i=1}^B \bar{x}_i = \frac{1}{B} \sum_{i=1}^B y_i$$

$$s_Y = \sqrt{\frac{1}{B-1} \sum_{i=1}^B (y_i - \bar{y})^2}$$

- Then increase n_{obs} and repeat

$j = 1$

$j = 2$

$j = 3$

$j = 4$

$j = 5$

$i = 1$











$\rightarrow y_1 = \frac{1}{n_{obs}} \sum_{j=1}^{n_{obs}} x_j$

$i = 2$











$\rightarrow y_2 = \frac{1}{n_{obs}} \sum_{j=1}^{n_{obs}} x_j$

$\vdots$

$i = B$









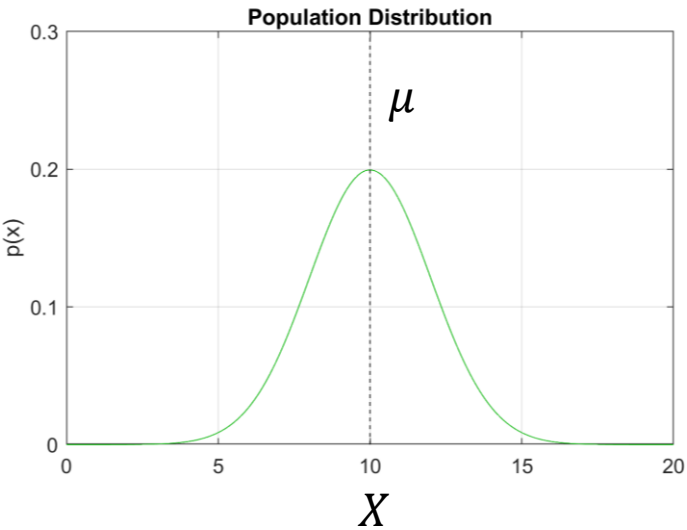


$\rightarrow y_B = \frac{1}{n_{obs}} \sum_{j=1}^{n_{obs}} x_j$

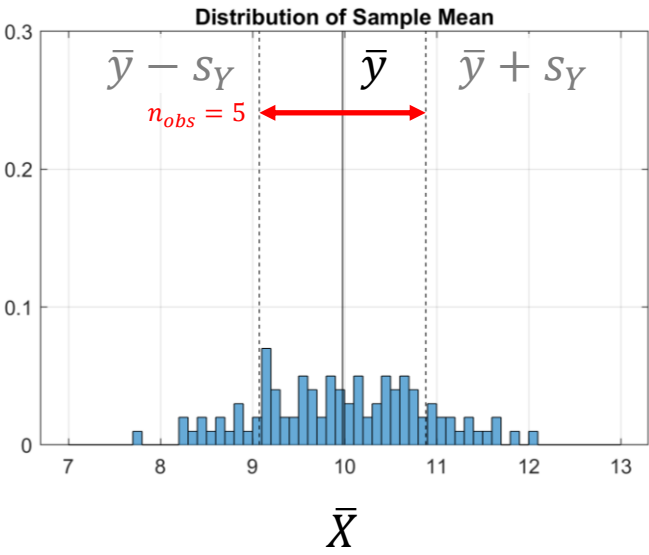
n_obs=5



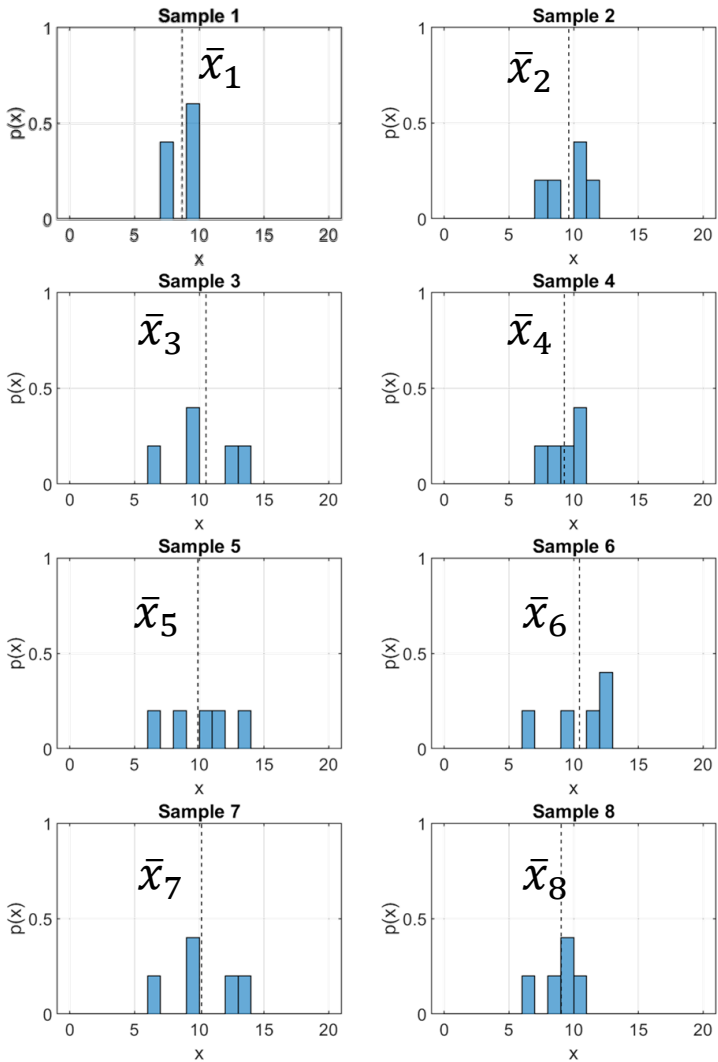
Population distribution



Distribution of sample means from 100 samples

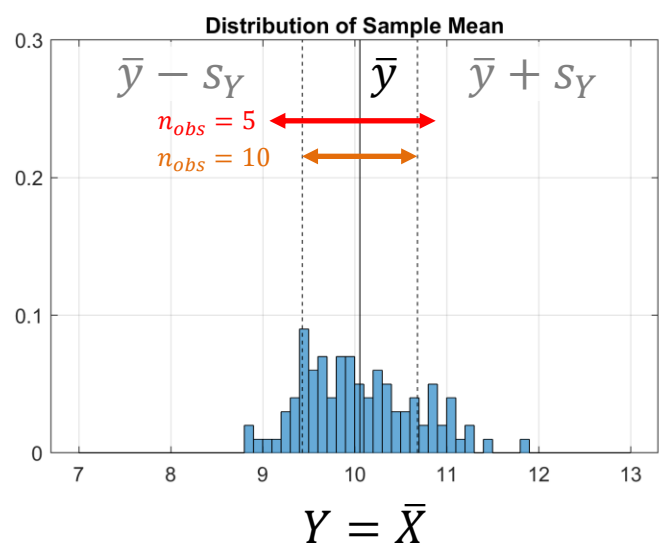
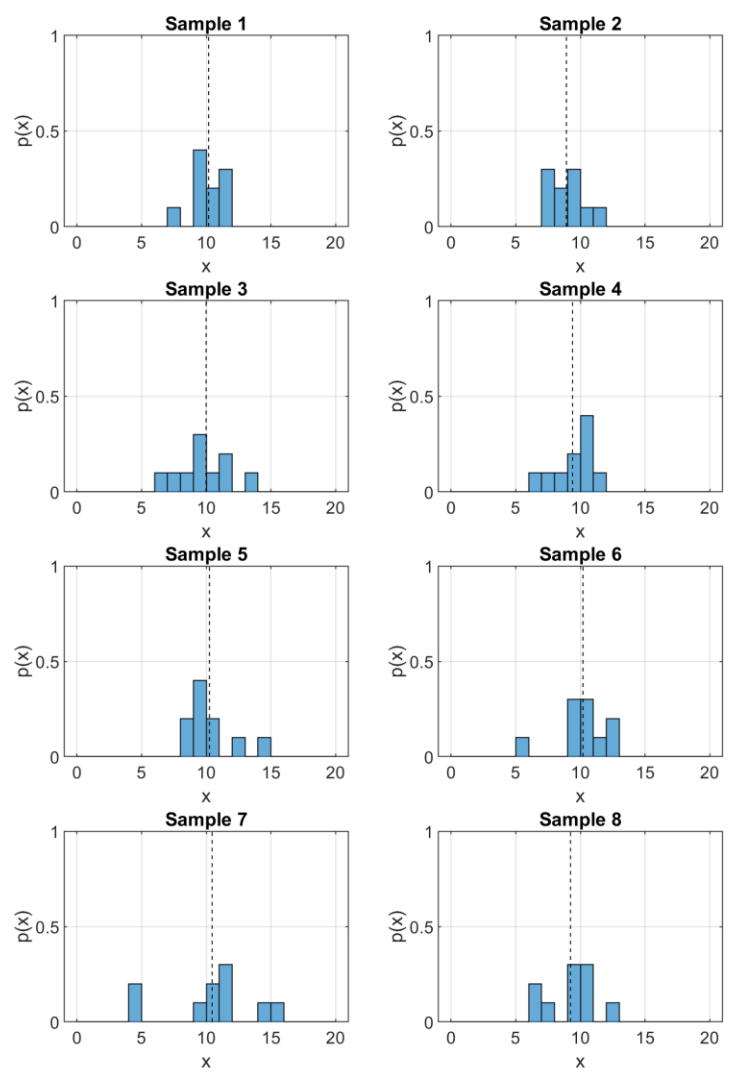
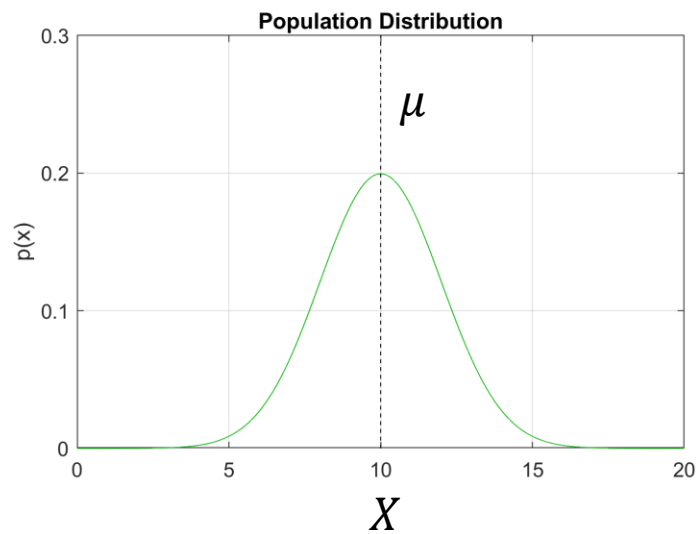


$$\bar{y} = \frac{1}{B} \sum_{i=1}^B \bar{x}_i$$

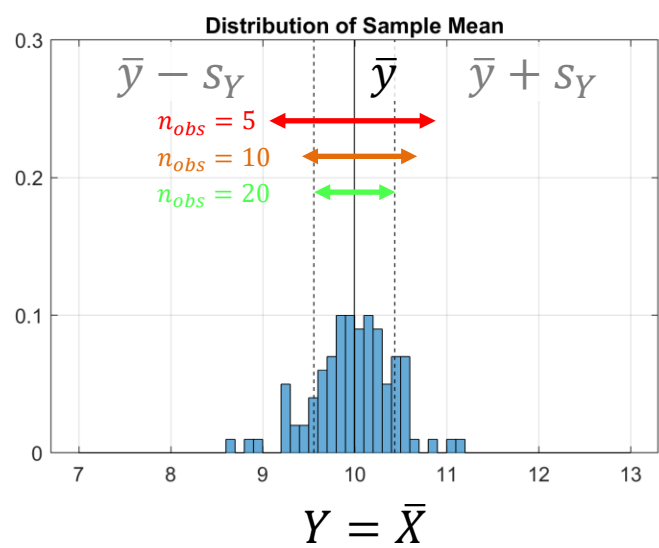
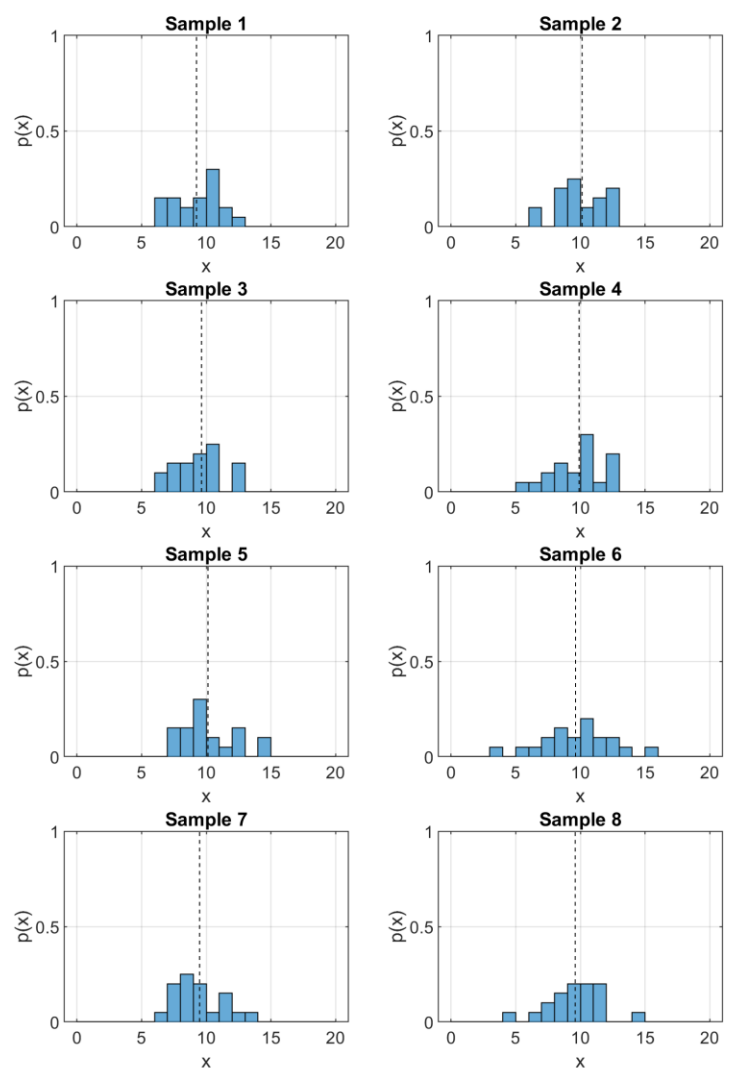
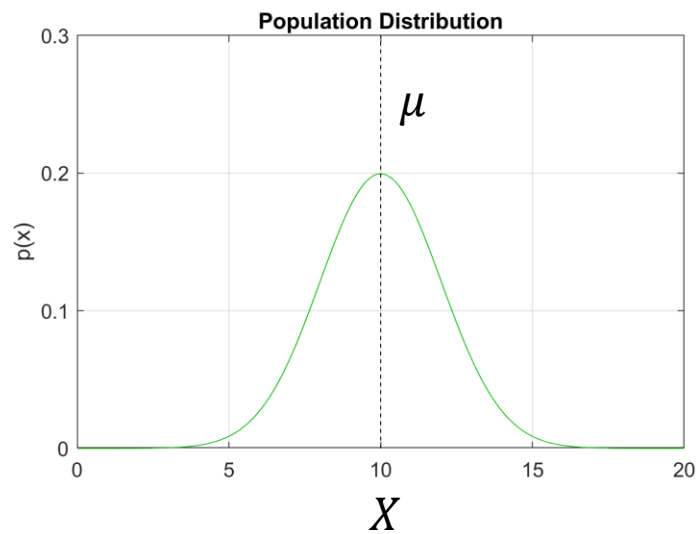


The first eight samples (out of 100)

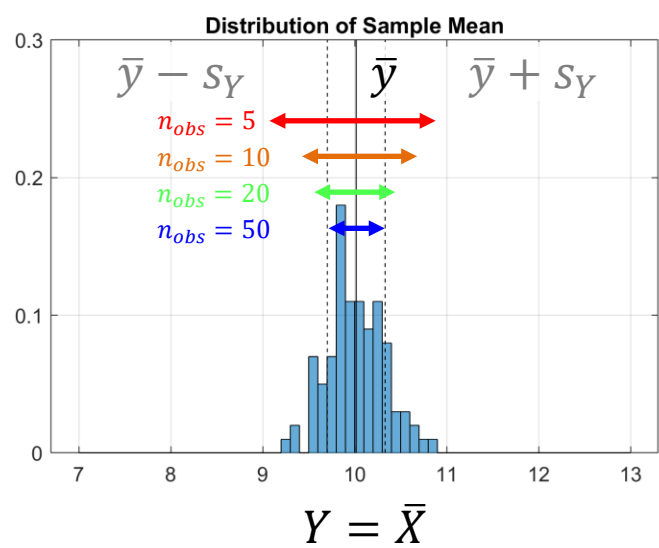
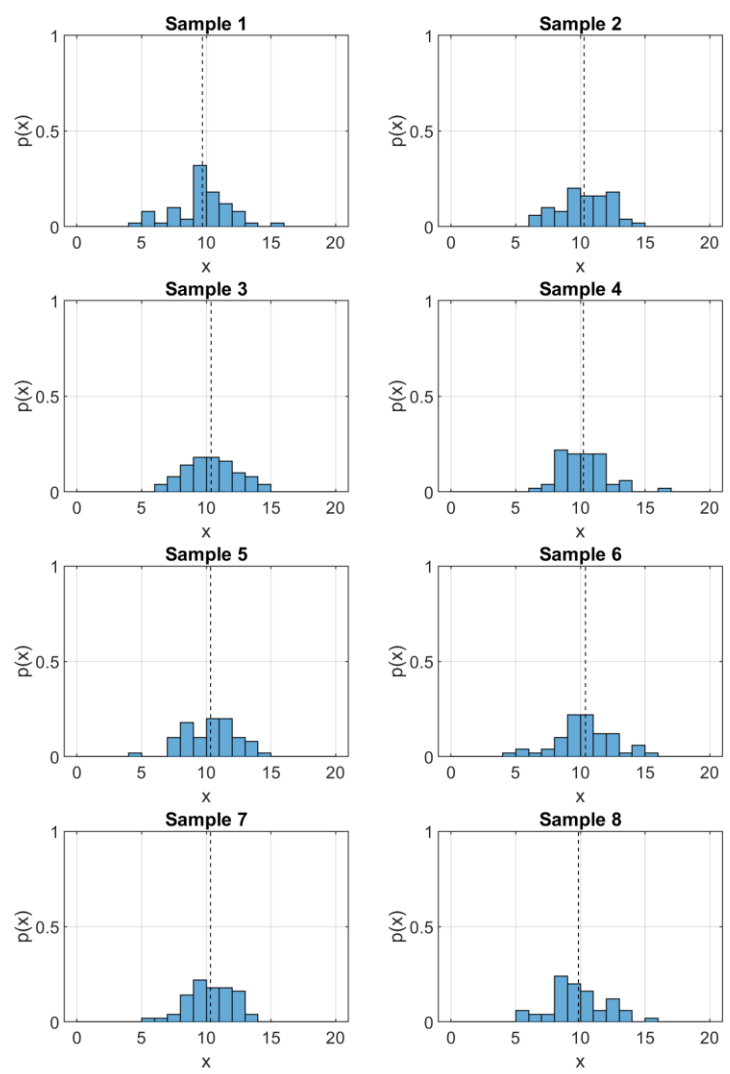
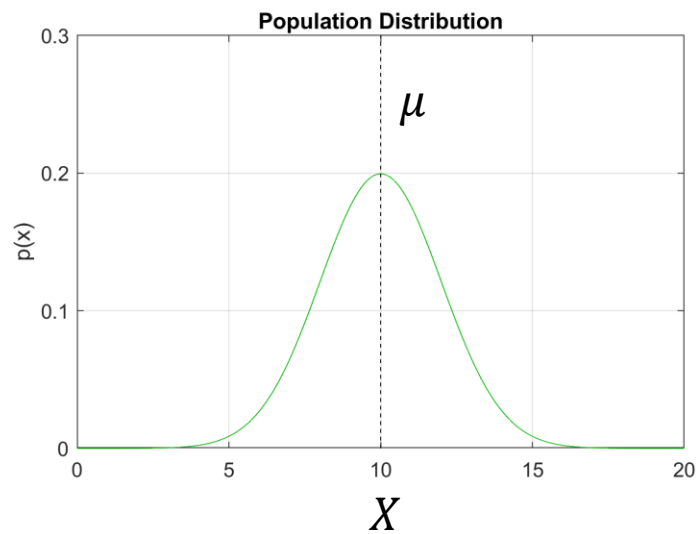
n_obs=10



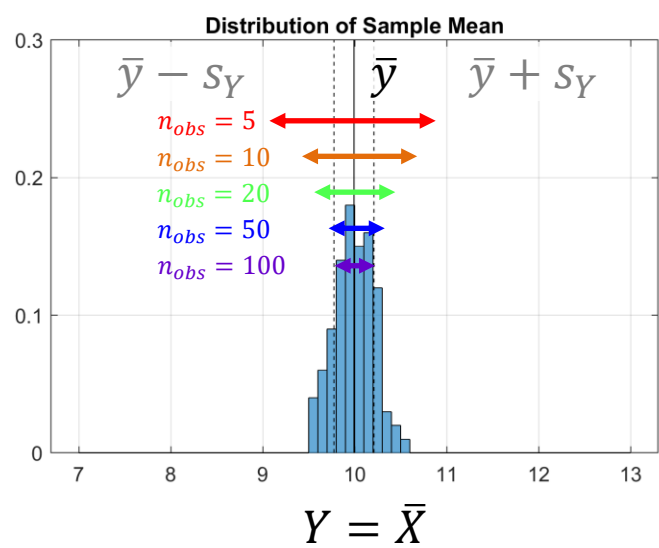
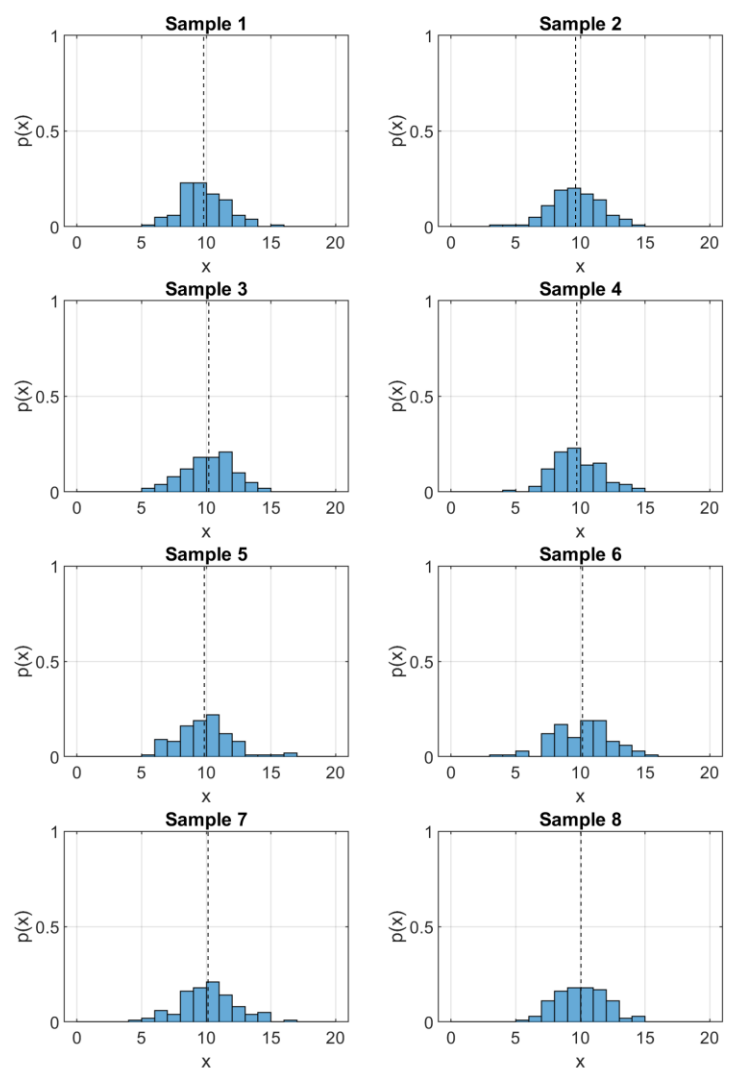
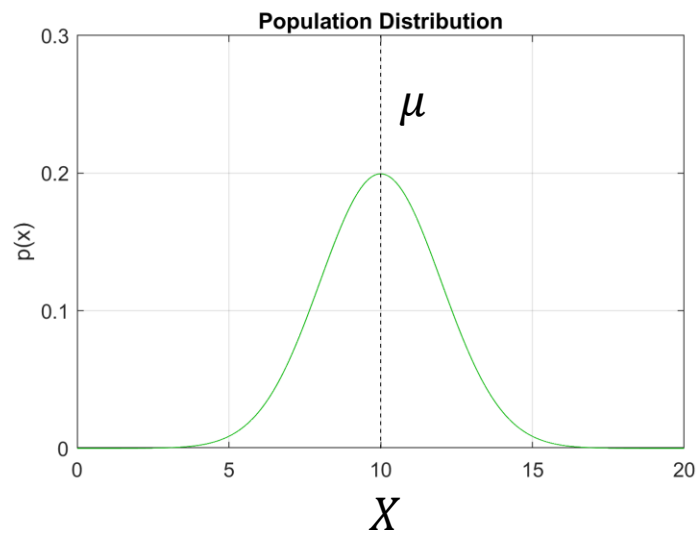
n_obs=20



nobs=50



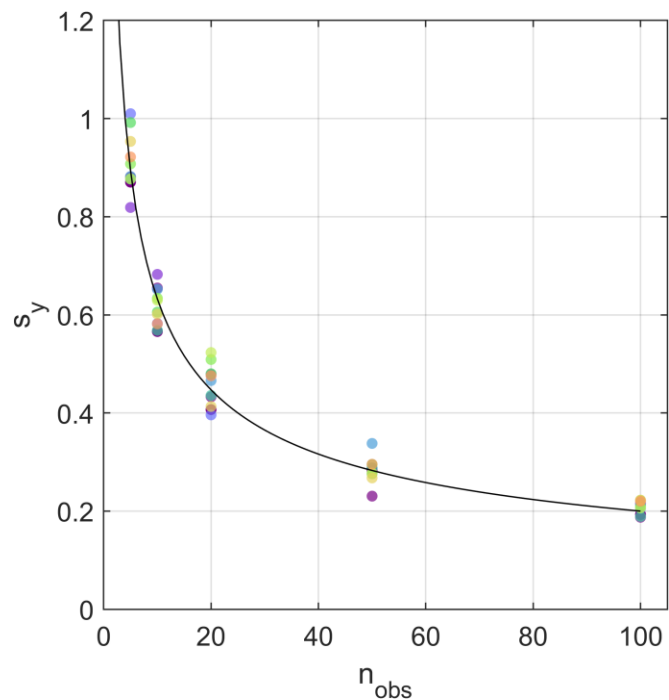
n_obs=100



What have we learned?

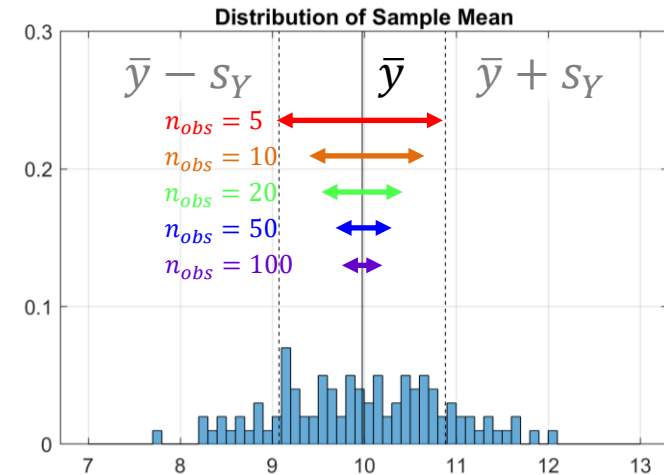


- As n_{obs} increased, the probability that a given $y = \bar{x}$ was close to μ increased
- The value of $\bar{y}$ was close to μ
- The value of $s_Y = s_{\bar{X}}$ decreased non-linearly as n_{obs} increased

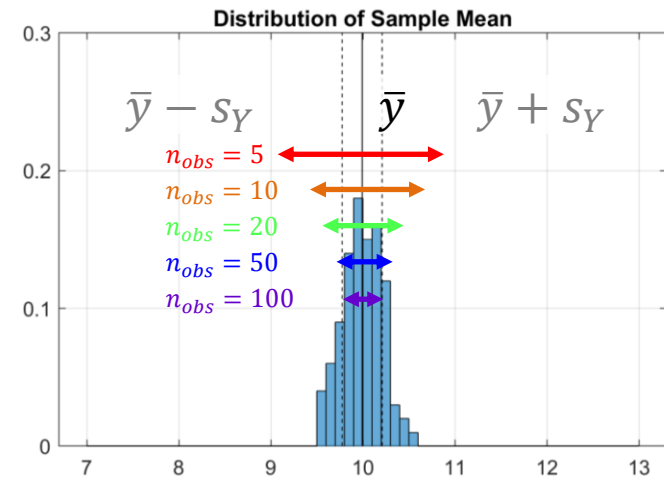


- Repeat 10 times for good measure
- Try a model where SD varies with square root of number of observations

$$S_Y = S_{\bar{X}} \approx \frac{\sigma}{\sqrt{n_{obs}}}$$



$n_{obs} = 5$



$n_{obs} = 100$

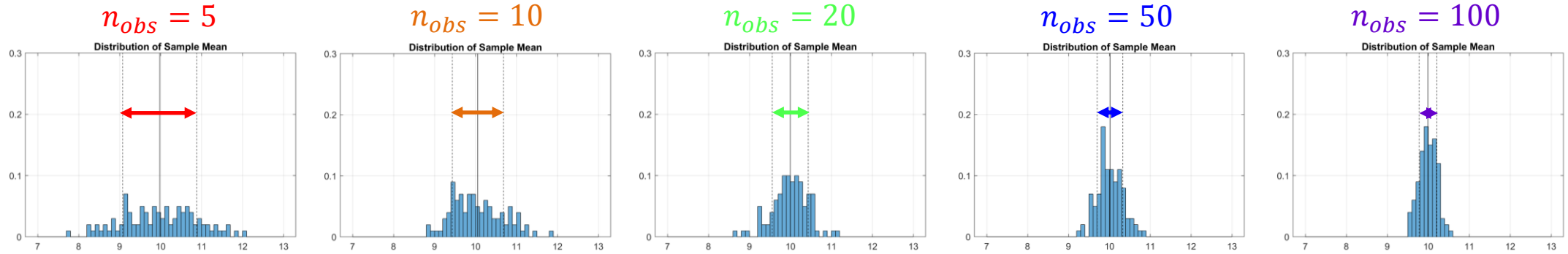


Central Limit Theorem

For large samples, the distribution of the sample means is normal

The Central Limit Theorem

For large samples, the distribution of the sample means is normal



- The mean of the sample means is approximately the population mean
- Standard deviation of the sample mean - **Standard Error of the Mean** – is inversely proportional to $\sqrt{n}$ $n = n_{obs}$
- **Irrespective of underlying distribution for large n – see Exercises**

For sufficiently large n

$$E[\bar{X}] = \mu_{\bar{X}} = \mu_X$$

$$V[\bar{X}] = \sigma_{\bar{X}}^2 = \frac{\sigma_X^2}{n}$$

$$\sigma_{\bar{X}} = \frac{\sigma_X}{\sqrt{n}}$$

Mean sample mean is the population mean

Variance of sample mean decreases as sample size increases

Standard error of the mean (SEM) is the standard deviation of sample means

A single foxes vs the badgers



designer.microsoft.com/image-creator

- A rival population of foxes has an average weight (X) of 6 ± 2 kg (mean $\pm$ SD).
- Only heavier foxes can compete with the badgers.
- i) If you selected a single fox, what is the probability that they weigh more than 8 kg?

$$X \sim N(\mu_X, \sigma_X^2) = N(6, 4)$$

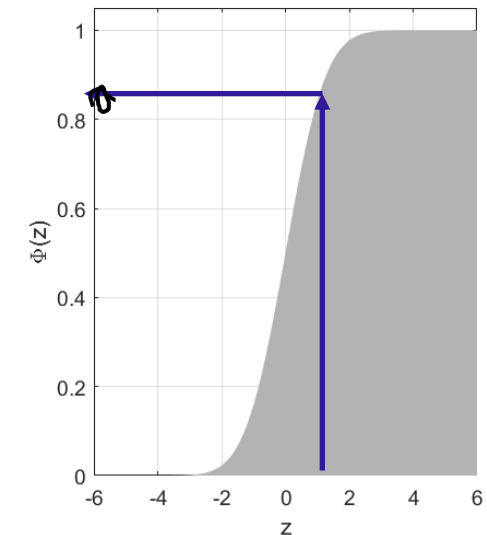
$$Z = \frac{(x - \mu_X)}{\sigma_X} = 1$$

$$P(X \geq 8) = P(Z \geq 1)$$

$$P(Z \geq 1) = 1 - \Phi[1.00]$$

$$= 1 - 0.8413 = 0.1587$$

	Second decimal place of z							
	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850

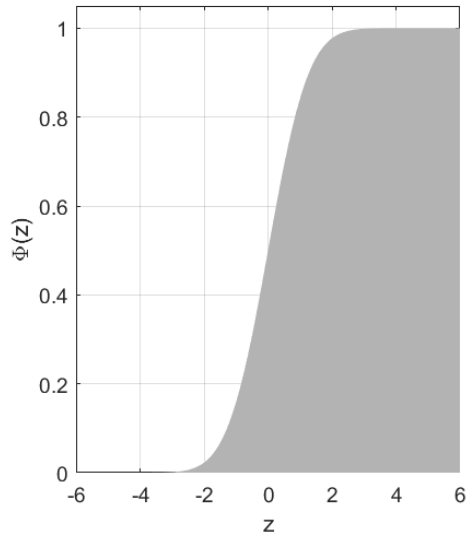


A skulk of foxes vs the badgers



- ii) If you collect a sample of 9 foxes, what is the probability that the sample average weight is more than 8kg?

$$X \sim N(\mu_X, \sigma_X^2) = N(6, 2^2)$$





Point Estimations

Accuracy and Precision

Point Estimators

- A point estimate of a parameter θ is a single number that can be regarded as a sensible estimate for θ
- A point estimate is obtained by selecting a suitable statistic and computing its value from the given sample data
- The selected statistic is called the point estimator $\hat{\theta}$ of θ 
- Examples of point estimators
 - Sample mean
 - Sample standard deviation
 - The maximum
 - The slope of a regression line

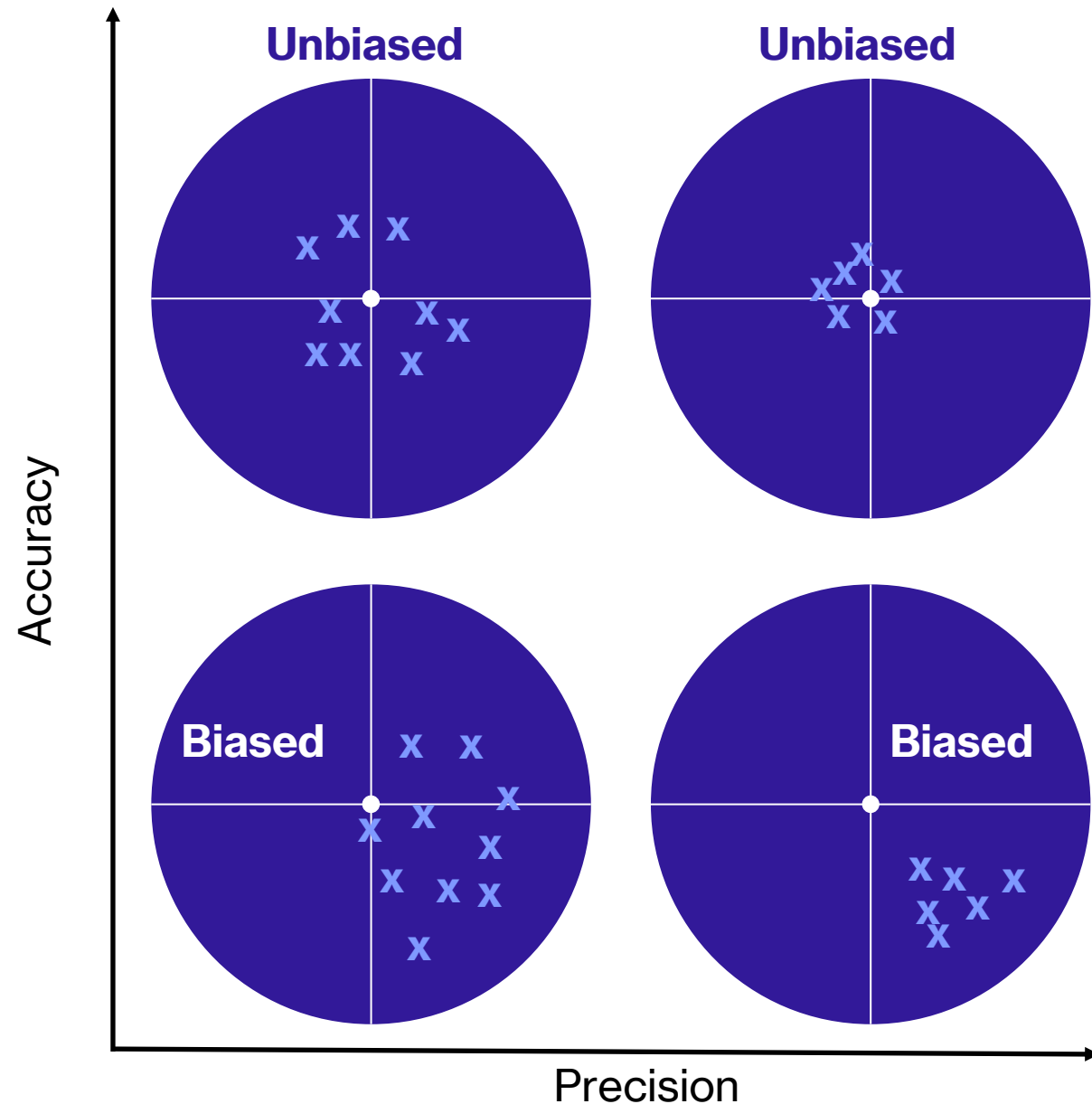
Height Point Estimators



- The height within the male population is described by $N \sim (\mu, \sigma)^2$ with $\mu = 178 \text{ cm}$ and $\sigma = 8 \text{ cm}$
 - Select 5 people and get $\{180, 187, 174, 176, 179\} \text{ cm}$
 - The sample mean $\bar{x}$ is an estimator of μ , such that $\bar{x} = \hat{\mu} = 179.2 \text{ cm}$
 - The sample median $\tilde{x}$ is another estimator of μ , such that $\tilde{x} = \hat{\mu} = 179 \text{ cm}$
 - Select 5 other people and get $\{181, 177, 175, 174, 178\} \text{ cm}$
 - The sample mean $\bar{x}$ is now $\bar{x} = \hat{\mu} = 177 \text{ cm}$
 - The sample median $\tilde{x}$ is now $\tilde{x} = \hat{\mu} = 177 \text{ cm}$
- Which is a better estimator?
 - In general, how do we choose how good an estimator is?

Accuracy and Precision

- Accuracy is how close a measure is to the true value
 - Analogous to systemic error or bias
 - Related to measure of location
- Precision is how close measures are to each other
 - Analogous to random error
 - Quantified with standard error – a measure of spread



Accuracy of Point Estimators

- The optimal estimator would be one with no error $\hat{\theta} = \theta$
- Generally, estimators will have an error, e_{θ} , such that

$$\hat{\theta} = \theta + e_{\theta}$$

- The expected value of $\hat{\theta}$ is therefore

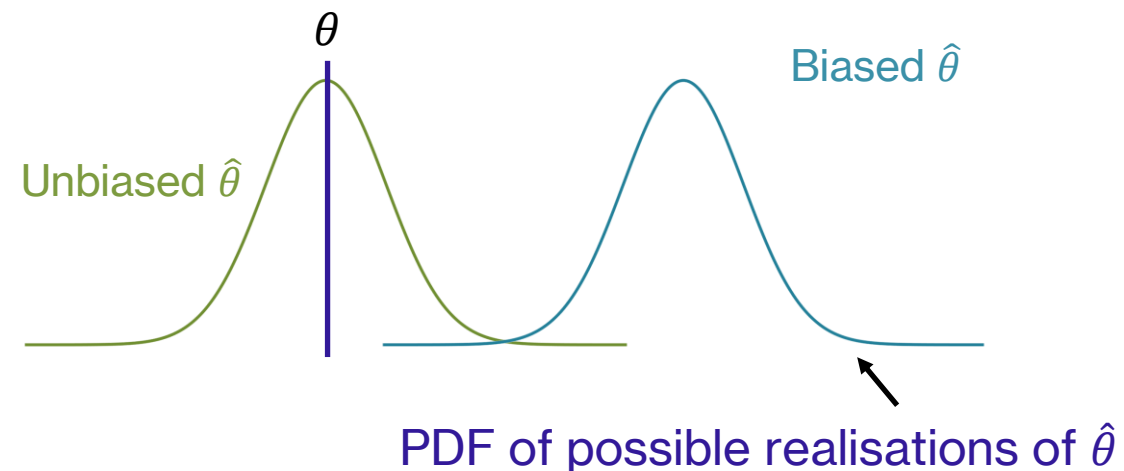
$$E(\hat{\theta}) = E(\theta + e_{\theta}) = E(\theta) + E(e_{\theta}) = \theta + E(e_{\theta})$$

As θ is an exact value

- If $E(e_{\theta}) = 0$ then on average the error tends to zero (irrespective of precision) – it is unbiased or accurate

$$\text{Bias} = e_{\theta} = E(\hat{\theta}) - \theta$$

- An estimator is biased if $E(\hat{\theta}) \neq \theta$
- **We need to know θ to quantify**

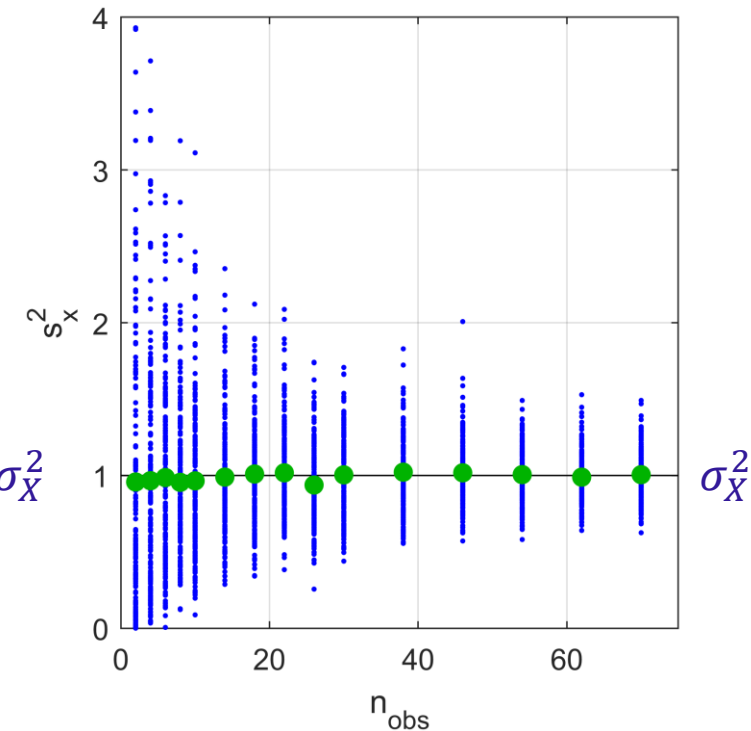
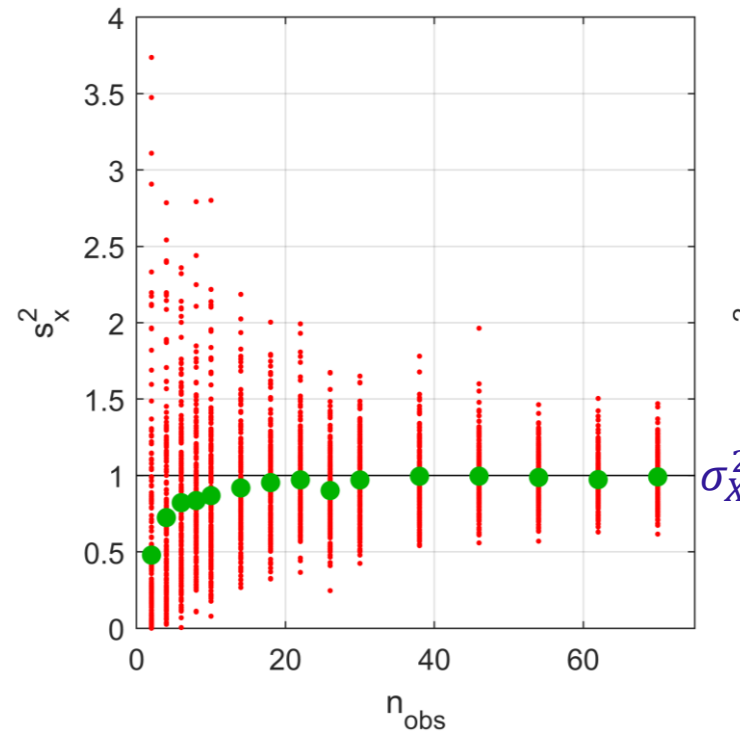


Bias of Variance

- Why do we use $n - 1$ when calculating variance?
- See `Lecture5_Code`
- $B = 200$ samples of n_{obs} observations
- $\sigma_X^2 = 1, \mu_X = 0$
- $\overline{s_X^2}$ is the average sample variance
- Without the ' $n - 1$ ', for low number of observations, the variance is an underestimate
 - The red s^2 is a biased predictor of the population variance
- With the ' $n - 1$ ', the variance is an unbiased
- But there is still not good precision!

$$s_X^2 = \frac{1}{n_{obs}} \sum_{i=1}^N (x_i - \mu)^2$$

$$s_X^2 = \frac{1}{n_{obs} - 1} \sum_{i=1}^N (x_i - \mu)^2$$



Precision of Point Estimators

- Consider our generic estimator

$$\hat{\theta} = \theta + e_{\theta}$$

- The variance $\hat{\theta}$ is therefore

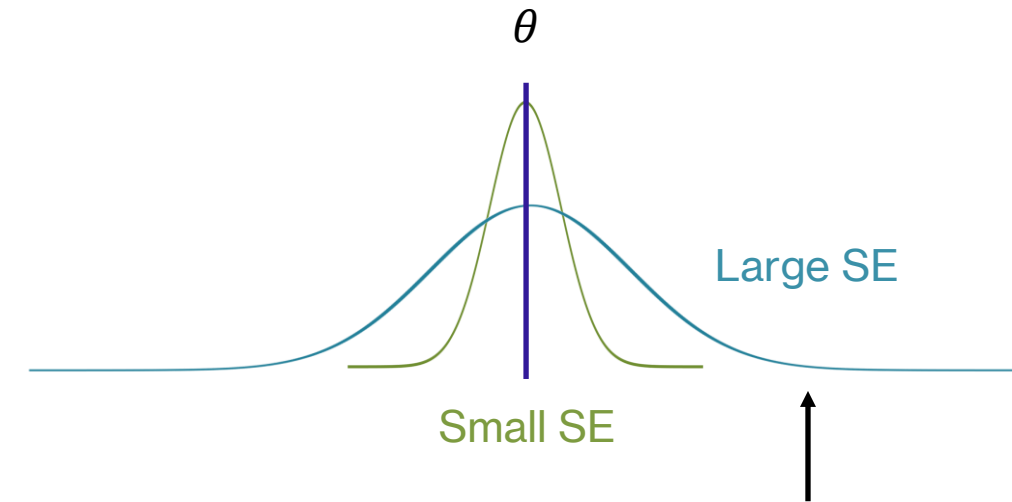
$$V(\hat{\theta}) = V(\theta + e_{\theta}) = V(\theta) + V(e_{\theta}) = V(e_{\theta})$$

As θ is an exact value

- The ‘precision’ of an estimator is termed the ‘standard error’

$$SE = \sqrt{V(\hat{\theta})}$$

- An estimator is precise if SE is small
- We can quantify for some known distributions
- Otherwise, bootstrapping simulations can be used



PDF of possible realisations of $\hat{\theta}$



Precision of the Sample Mean

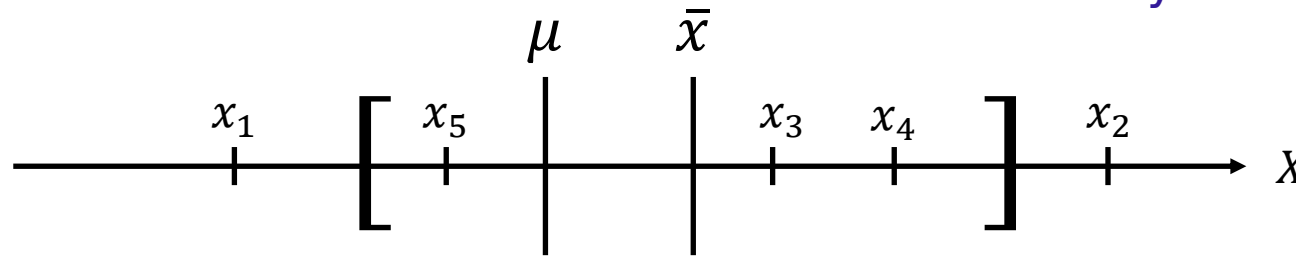
Standard error, Margin of Error and Confidence Intervals

Barley Yield



- Guinness wanted to know the type of barley that produced a maximal yield in ton/hectare (t/ha)
- Suppose we have data from 5 years, with RV X representing the yield in a given year

year=observation



- The sample mean $\bar{X}$ is an estimator of the population mean μ
- Without knowing μ , we cannot evaluate accuracy
- We can ask ‘how precise is a given $\bar{x}$?’
- We quantify this using Confidence Intervals



<https://www.guinness.com/en-us/>



Standard Error of the Mean

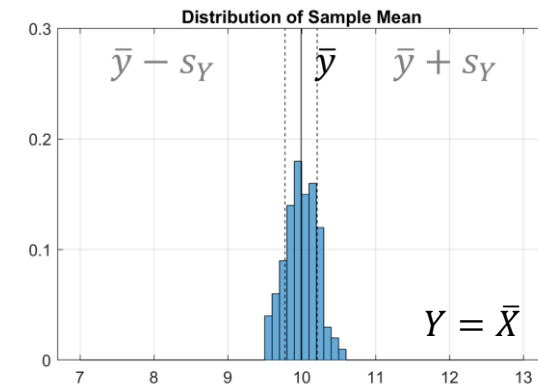
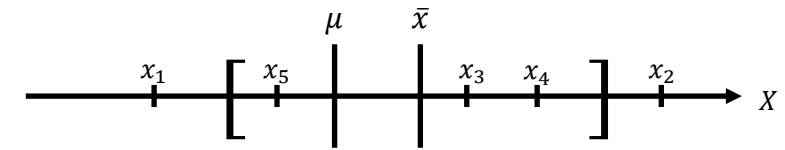
- $\bar{X}$ is a point estimator of μ
- It is unbiased, as $E(\bar{X}) = \mu$
- From the CLT simulations we learned that it has standard error given by

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$$

- This is the standard error of the mean (SEM)
- How can we interpret $\sigma_{\bar{X}}$?
- Consider the interval

$$[\bar{x} - \sigma_{\bar{X}}, \bar{x} + \sigma_{\bar{X}}]$$

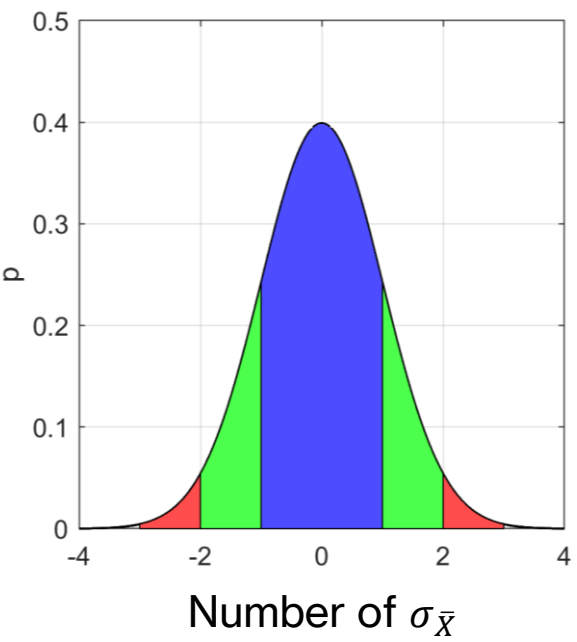
- Going back to our CLT simulations, we can look at how many of our simulations were in the range $[\bar{x} - \sigma_{\bar{X}}, \bar{x} + \sigma_{\bar{X}}] \sim [y - s_y, y + s_y]$
- We would find that approximately 68 out of 100 of our sample means would be in this range
- SEM, $\sigma_{\bar{X}}$, a 68% **'margin of error'**
- $[\bar{x} - \sigma_{\bar{X}}, \bar{x} + \sigma_{\bar{X}}]$ is 68% **'confidence interval'**



we'll address the σ vs s question next week

Margin of Error

- The range $[\bar{x} - \sigma_{\bar{x}}, \bar{x} + \sigma_{\bar{x}}]$ is our 68% confidence interval, but 68% feels a bit arbitrary and not that useful
- More useful might be 95% of the data or 99% of the data
- Hence, instead of one $\sigma_{\bar{x}}$ we can use a multiple to get the $100(1-\alpha)\%$ Margin of Error, ME:



$$100(1 - \alpha)\% ME_{CI} = z_{1-\alpha/2}\sigma_{\bar{x}}$$

• Where

68% ME

$\alpha = 0.32$

$1 - \frac{\alpha}{2} = 0.84$

$z_{1-\frac{\alpha}{2}} = \Phi^{-1}(0.84) = 1.0$

68% ME = 1.0 $\sigma_{\bar{x}}$

95% ME

$\alpha = 0.05$

$1 - \frac{\alpha}{2} = 0.975$

$z_{1-\frac{\alpha}{2}} = \Phi^{-1}(0.975) = 1.96$

95% ME = 1.96 $\sigma_{\bar{x}}$

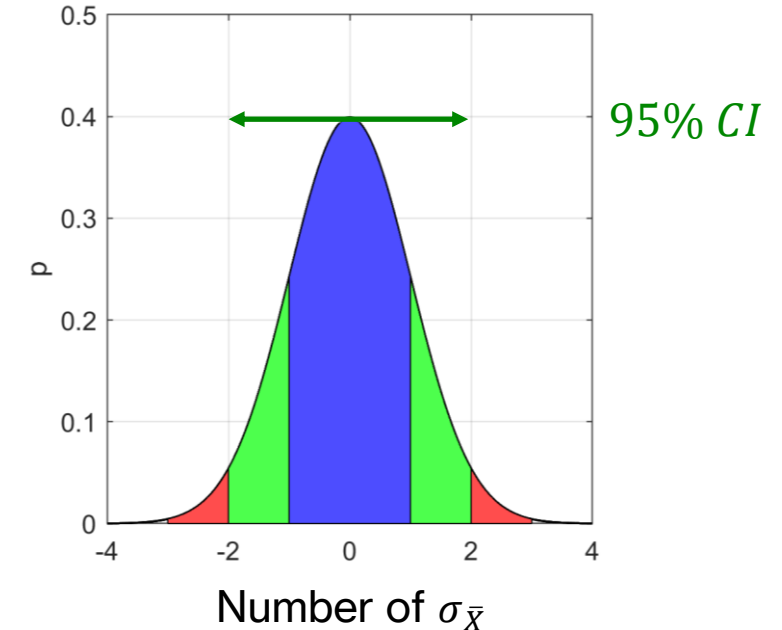
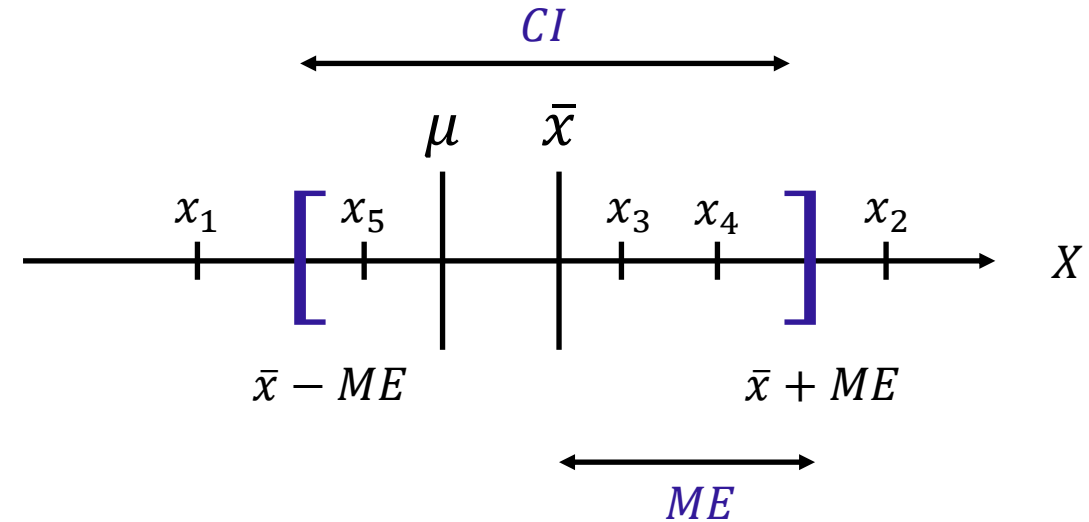
		Second decimal place of z									
		0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359	
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753	
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141	
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517	
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879	
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224	
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549	
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852	
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133	
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389	
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621	
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830	
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015	
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177	
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319	
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441	
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545	
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633	
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706	
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767	
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817	
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857	
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890	
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916	
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936	
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952	
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964	
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974	
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981	
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986	
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990	
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9992	0.9993	
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995	
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997	
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998	

Confidence Intervals

- The range or interval covered by the mean $\pm$ the margin of error

$$CI = [\bar{X} - ME_{CI}, \bar{X} + ME_{CI}]$$

- Common (imprecise) interpretation
 - The probability that μ lies within a given 95% CI is ~95%
 - $P(\bar{x} - ME \leq \mu \leq \bar{x} + ME) = 0.95$
 - Technically incorrect because, once you have a value for $\bar{x}$ it is no longer a statistic, so we cannot define an associated probability
- Proper interpretation
 - If we took 100 samples, 95 of the CIs would contain μ
 - Frequentist probability
- An 'intuitive' indication of the precision of our estimate of the mean



FYI - Confidence Intervals Derivation

- Standardise the sample mean $\bar{X}$ by the population mean and the standard error on the mean using $\sigma_{\bar{X}}$

Added the subscript for clarity

$$Z_{\bar{X}} = \frac{\bar{X} - \mu}{\sigma_{\bar{X}}}$$

- For 95% CI, there is a 95% probability that the standardised value $Z_{\bar{X}}$ is in the range $[-1.96, +1.96]$ (see figure)

$$P\left(-1.96 < \frac{\bar{X} - \mu}{\sigma_{\bar{X}}} < 1.96\right) = 0.95$$

Multiply out SE

$$-1.96\sigma_{\bar{X}} < \bar{X} - \mu < 1.96\sigma_{\bar{X}}$$

Subtract $\bar{X}$

$$-\bar{X} - 1.96\sigma_{\bar{X}} < -\mu < -\bar{X} + 1.96\sigma_{\bar{X}}$$

Multiply by -1

$$\bar{X} + 1.96\sigma_{\bar{X}} > \mu > \bar{X} - 1.96\sigma_{\bar{X}}$$

Flip it round

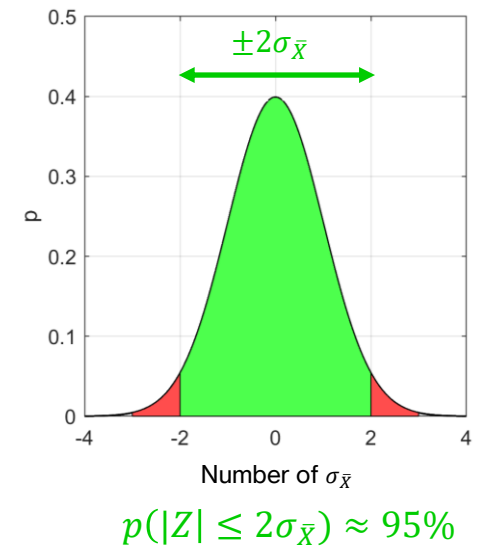
$$\bar{X} - 1.96\sigma_{\bar{X}} < \mu < \bar{X} + 1.96\sigma_{\bar{X}}$$

- Generally

$$P(\bar{X} - z_{1-\alpha/2}\sigma_{\bar{X}} < \mu < \bar{X} + z_{1-\alpha/2}\sigma_{\bar{X}}) = 1 - \alpha$$

“The probability that μ is in the $(1 - \alpha) \times 100\%$ confidence interval is $1 - \alpha$ ”

- Note derivation all for RV $\bar{X}$, rather than specific realisation $\bar{x}$



Barley Yield Confidence Interval



- From 5 observations, we find that

$$\bar{x} = 5.4 \frac{t}{ha}$$

- We assume we know the population SD

$$\sigma = 0.1 \frac{t}{ha}$$

- What is the 90% confidence interval

- We can therefore report that
 - ‘the average barley yield was 5.40 [5.33, 5.47] t/ha’
 - ‘the average barley yield was 5.40 ± 0.07 t/ha’

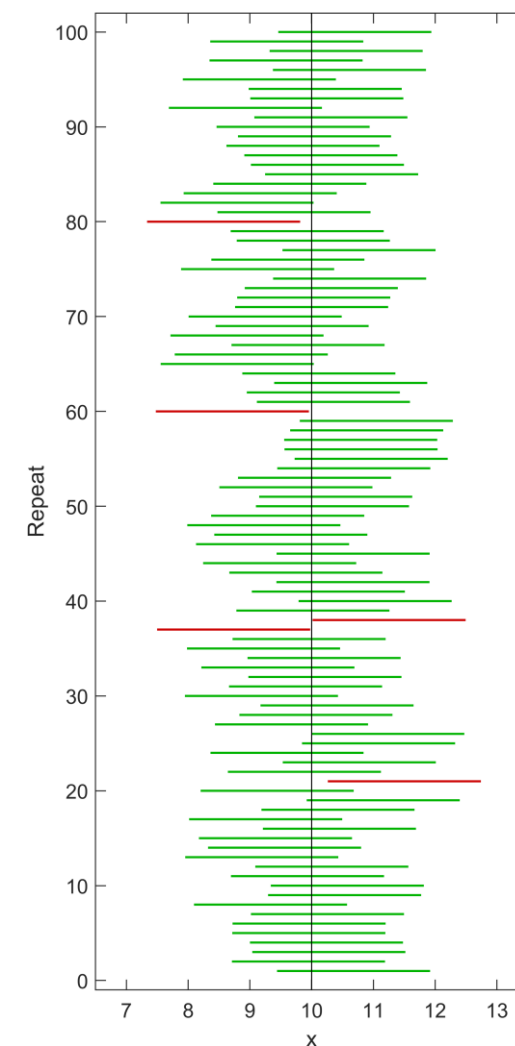
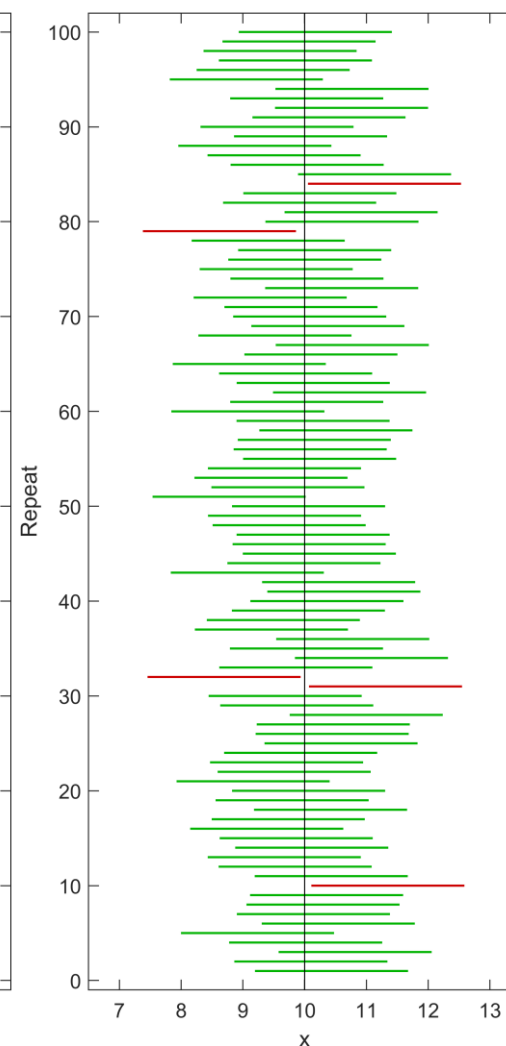
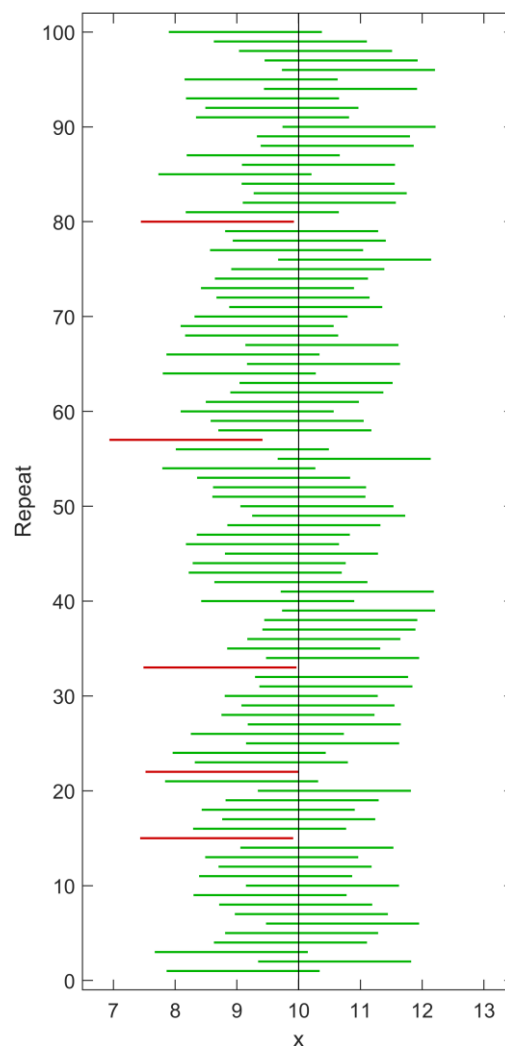
		Second decimal place of z						
		0.00	0.01	0.02	0.03	0.04	0.05	0.06
First decimal place of z	0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239
	0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636
	0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026
	0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406
	0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772
	0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123
	0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454
	0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764
	0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051
	0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315
	1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554
	1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770
	1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962
	1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131
	1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279
	1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406
	1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515
	1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608
	1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686
	1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750
	2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803
	2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846
	2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881



CI for badger data

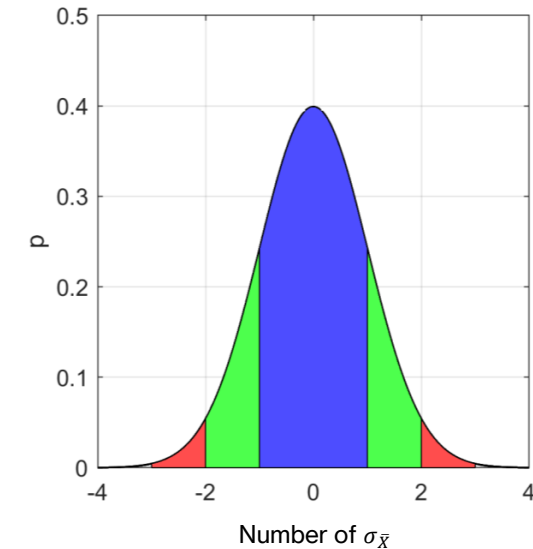


- Using $n_{\text{obs}}=10$ and $B=100$
- For each sample, plot confidence interval
- Colour red if it doesn't include μ
- Run a few times to demonstrate sampling



ME_{CI} will increase if we have...

- A smaller α
 - Larger interval to increase probability of incorporating the average
 - E.g. 95%: $z_{0.975} = 1.96$, 99%: $z_{0.995} = 2.58$
- More variability of underlying distribution
 - Larger σ , hence larger ME
- Smaller sample size
 - Will increase SE and hence ME
 - Important in experimental design
- In practice, you should (almost) **always** report a mean value along with its uncertainty (ME or CI)



$$ME = Z_{1-\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}$$

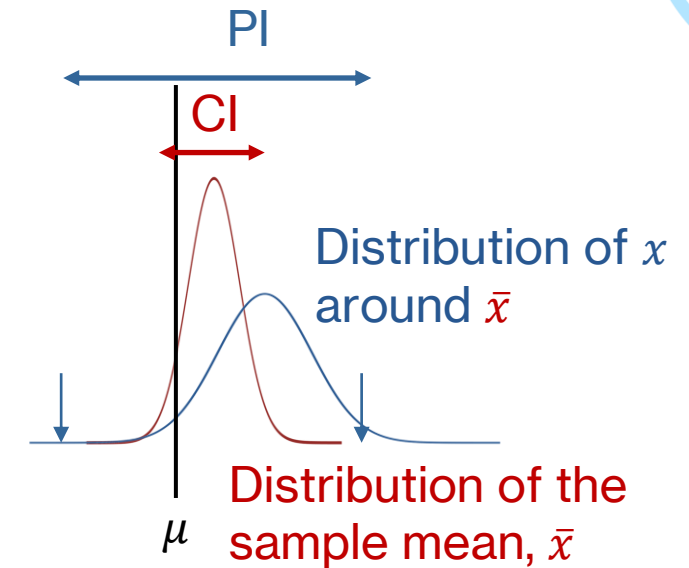


Precision of a future value

Prediction intervals

Prediction intervals

- Confidence interval tells us only about the mean, $\bar{X}$
 - They tell us about **uncertainty**
 - They tell us about a previous sample
- If we are interested in predicting a future value of X we use Prediction Intervals
 - They tell us about **variability**
 - They tell us about a future observation
- We have to combine
 - uncertainty on the mean (σ^2/n)
 - variability around the mean (σ^2)
- Recalling that we add variances



$$ME_{CI} = z_{1-\frac{\alpha}{2}} \sigma_{\bar{X}} = z_{1-\frac{\alpha}{2}} \sqrt{\sigma^2/n}$$

$$ME_{PI} = z_{1-\frac{\alpha}{2}} \sqrt{\sigma^2/n + \sigma^2}$$

Note, this isn't standard terminology, but I invented it because it's efficient for the reference sheet and aids understanding

Prediction intervals



$$ME_{PI} = z_{1-\frac{\alpha}{2}} \sqrt{\sigma^2/n + \sigma^2}$$

- Prediction intervals are used relatively uncommonly
- They should be used a lot more commonly!
- For our barley yield question, if we were to take a sixth observation, what range would we expect it to be in?
- What is the 95% prediction interval on a future observation?

$$100(1 - \alpha)\% \quad ME_{PI} = z_{1-\frac{\alpha}{2}} \sigma_X \sqrt{1 + 1/n}$$

$$PI = [\bar{X} - ME_{PI}, \bar{X} + ME_{PI}]$$

$$\bar{x} = 5.4 \frac{t}{ha}$$

$$\sigma = 0.1 \frac{t}{ha}$$

FYI - Prediction Intervals Derivation

- We took n samples to calculate $\bar{X}$
- The point estimator of the next observation X_{n+1} would be the mean, $\bar{X}$
- There will be a corresponding error

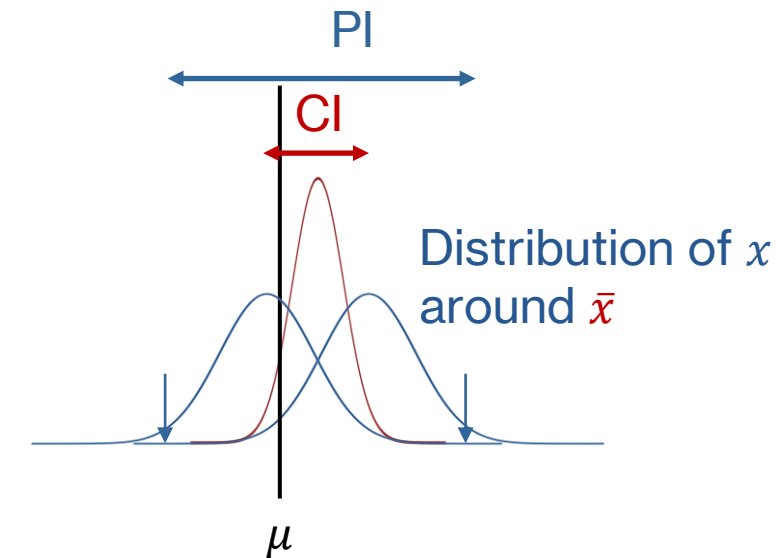
$$\bar{X} - X_{n+1}$$

- The expected value of the error is

$$E(\bar{X} - X_{n+1}) = E(\bar{X}) - E(X_{n+1}) = \mu - \mu = 0$$

- The variance of the error would be

$$\begin{aligned} V(\bar{X} - X_{n+1}) &= V(\bar{X}) + V(X_{n+1}) \\ &= \frac{\sigma^2}{n} + \sigma^2 = \sigma^2 \left(\frac{1}{n} + 1 \right) \end{aligned}$$



Normal Distribution, Statistics and Intervals

What we covered

- Normal Distribution
 - Standard Normal Distribution
 - Numbers of sigma
- Definition of a Statistic
- Central Limit Theorem
- Point Estimators
- Precision of the sample mean
 - SE, ME and CI
- Precision of a future value
 - PI

Up next

Lecture 6: Introduction to NHST